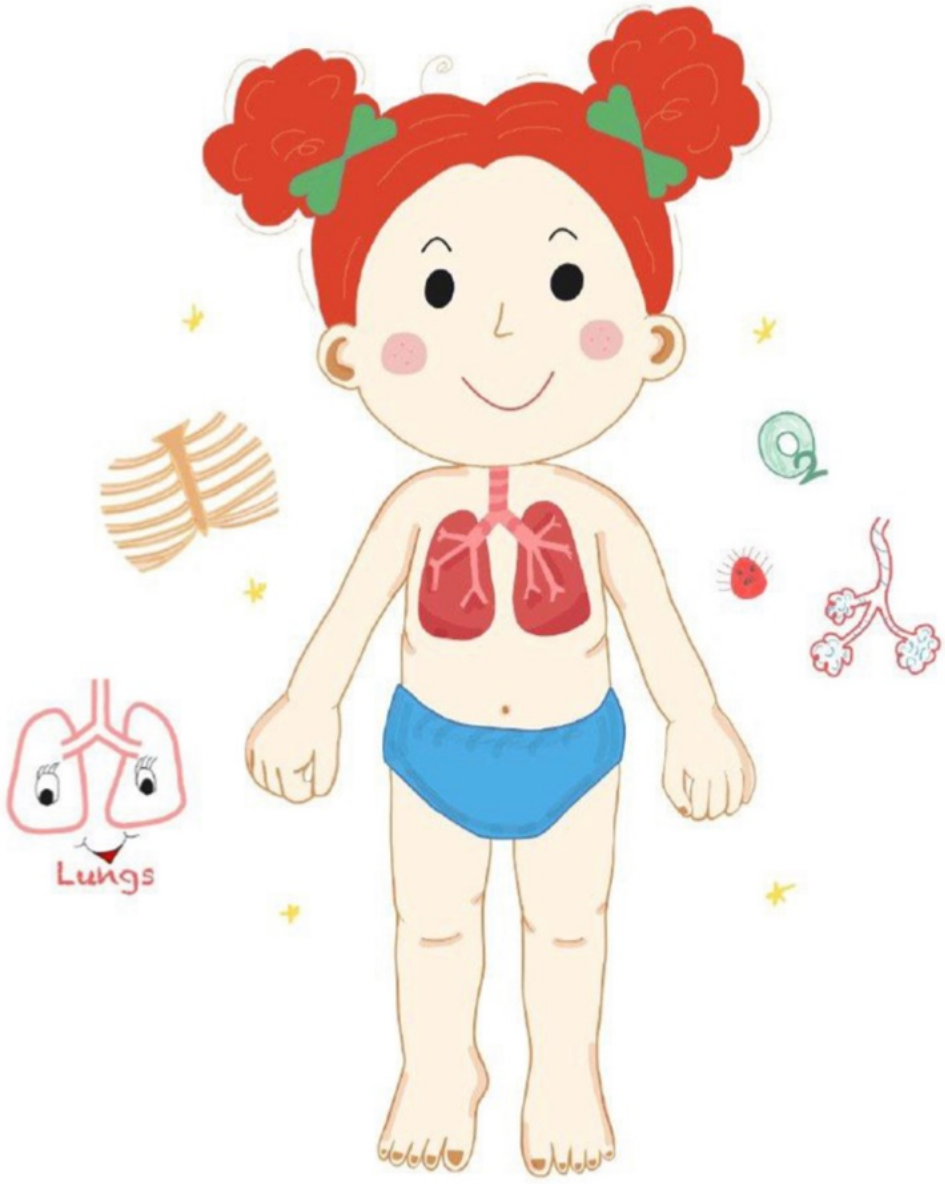
PEDIATRIC RESPIRATORY SYSTEM



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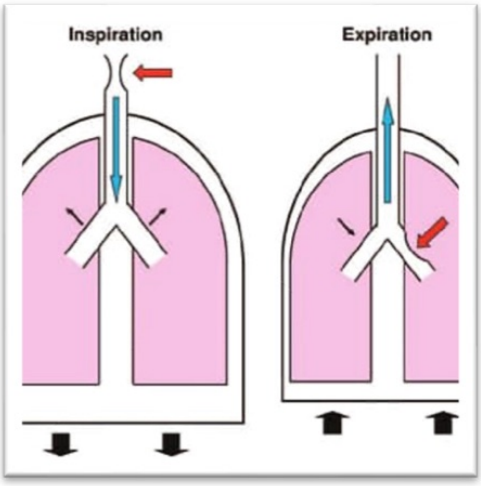
CONTENTS

1. Introduction: Symptoms and signs of respiratory system
2. noisy breathing and sign of RD
3. Respiratory tract infection: Pharyngitis, otitis media and sinusitis
4. Stridor: Laryngomalacia, croup and acute epiglottitis
5. Acute bronchiolitis
6. Pneumonia
7. Foreign body aspiration
8. Bronchial asthma
9. Cystic fibrosis
10. Bronchiectasis
11. Primary ciliary dyskinesia
12. Pleural effusion

Introduction

The respiratory system divided into:–



Upper airway "URT"

- Above the thoracic inlet
- The **upper** airway during breathing (**narrowed** during inspiration and **dilated** during expiration)

Lower airway "LRT"

- Below the thoracic inlet
- The **lower** airways during breathing (**dilated** during inspiration and **narrowed** during expiration)

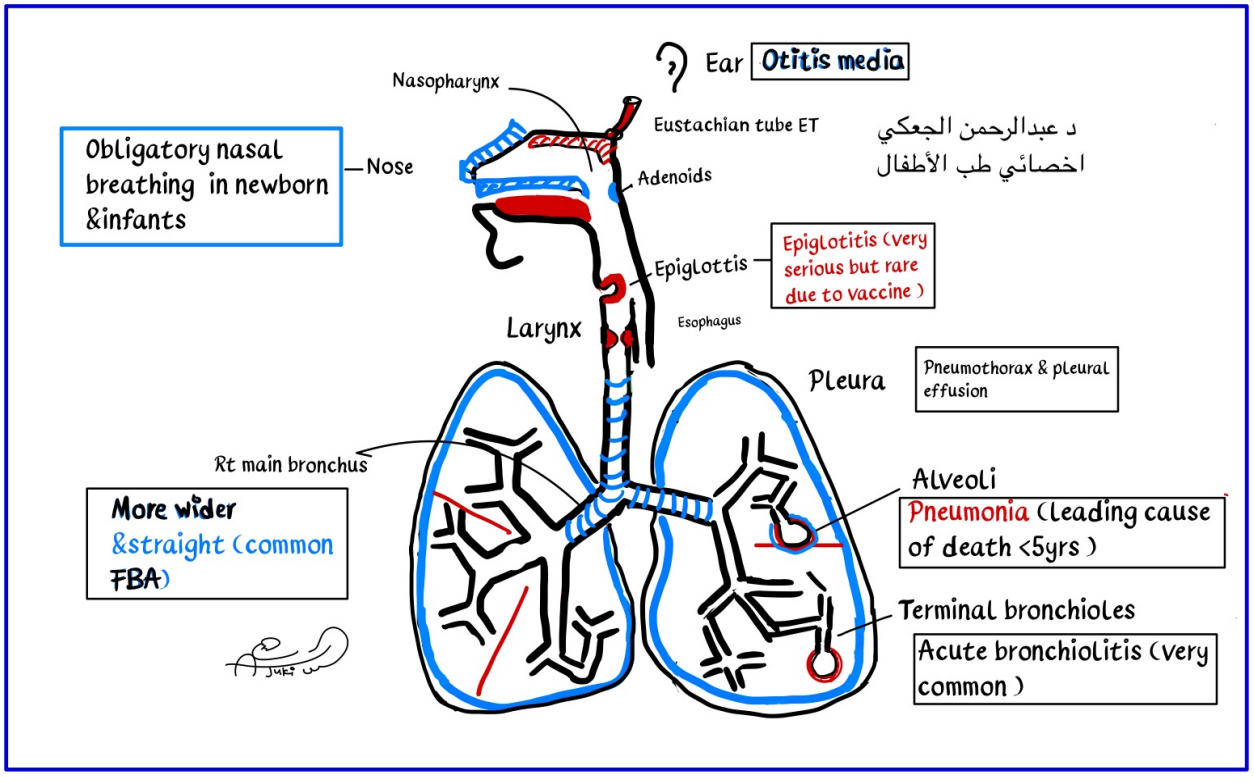
- so during airways obstruction the abnormal noisy sounds that created are in theU pper airways obstruction –>(stridor) heard mainly during inspiration
- Lower airways obstruction –>(wheeze) heard mainly during expiration

NOTE

a. The upper airways or upper respiratory tract includes the nose and nasal passages, paranasal sinuses, the pharynx, and the portion of the larynx above the vocal folds (cords).

b. The lower airways or lower respiratory tract includes the portion of the larynx below the vocal folds, trachea, bronchi and bronchioles.

c. infants & very young children have smaller airways and their lungs are growing so (the younger age hase more sever disease especially lower respiratory infection)



Symptoms and Sign of Respiratory disease:

1. Breathlessness (Increase work of breathing)

Can be due to

- Obstruction of airways e.g. Asthma
- Disease in the lung parenchyma e.g. Pneumonia
- Disease in the pleura e.g. Pneumothorax
- Extra pulmonary diseases as cvs e.g. Heart failure /or blood e.g anemia & metabolic acidosis.

2. Cough

(Is protective response to prevents invasion of lungs by microorganisms and for removal of secretion)

Types: (**wet (moist)** cough and **dry** cough) / (**acute** and **chronic** cough).

The characteristics of cough	Associated disease
Spasmodic whooping cough	Pertussis
Barking (brassy) cough	Croup
Staccato cough	Chlamydia infection
Sudden cough after chocking episodes	Foreign body aspiration
Night time (nocturnal) cough	Br. asthma
Morning cough	Sinusitis, allergic rhinitis
Recurrent coughing with exercise	Exercise induced asthma
Early morning cough	Posterion nasal drip

Chronic cough: Is defined as daily cough lasting longer than 4 weeks (>4–6 weeks)

Causes:

- Bronchial asthma and allergic rhinitis
- TB – missed FBA "if not removed"
- Cystic fibrosis and bronchiectasis
- GERD
- Psychogenic in origin

3. Noisy breathing:

Audible sound usually caused by a blockage in the air passages, is common condition in child include:

a. **Stridor**: Is audible, harsh, high pitched noisy sound due to partial obstruction in the upper airways heard mainly during inspiration.
b. **Wheeze**: Is audible, musical sound due to passage of air in narrow lower airways heard mainly during expiration.

c. **Grunting**: Is expiratory sound due to forced expiration against partially closed glottis "its sign of respiratory distress RD". as in bronchiolitis –respiratory distress syndrome–pneumonia

d. **Snoring**: Is a sound generated during sleep by vibration of loose tissue in the upper airways and caused by (obesity, down syndrome, adenoid hypertrophy and tonsillar hypertrophy SCA), adenoids hypertrophy may leads to sleep apnea syndrome.

e. **Stertor**. It is typically low–pitched, created in the nose or the back of the throat, most closely sounds like nasal congestion & associated with common cold, or like the sound made with snoring,

Signs of Respiratory distress:

1. **Tachypnea**: Is increase in respiratory rate "RR" above certain limit according to age

- Birth – 2 months. (> 60 Breath/min)
- 2 months – 1 year (> 50 Breath/min)
- 1 year – 5 years. (> 40 Breath/min)
- 6 years – 9 years (>30 Breath/min)

Table 2.1 Normal respiratory rate and tachypnoea by age group

Normal	Breaths/min	Tachypnoea
Neonate	35–60	>60
Infant	30–40	>50
Young Child	20–30	>40
School-age child	20–25	>30
Adolescent	15–20	>30

Tachypnoea is one of the most sensitive signs of respiratory illness. **However**, this is not specific for respiratory problems and also occurs in heart failure, metabolic disorders and sepsis

2. **Nasal flaring (acting alae nasi)**

3. **Chest recession "Retraction"**

- Subcostal recession "SCR"
- Intercostal recession "ICR"
- Supra sternal recession "SSR"
- Sternal recession (in newborn)

4. **Grunting**

5. **Cyanosis**

Grades of RD

I = mild = Tachypnea + nasal flaring
II = Moderate = + Recession
III = Sever = + Grunting
IV = Advanced = + cyanosis and disturbed conscious level

Respiratory disorders

- **Commonest illness** in children (they account about 50% of all illness < 5 years and 30% of all illness 5–12 years of ER or OPD visit)
- It constitute about **30%** of hospital **admission**
- Br **asthma** is the **most common chronic illness** of childhood in the UK

Risk factors for respiratory diseases in children:

- Poor socioeconomic status,overcrowding.
- Large family size.
- Parental genetic predisposition and smoking (maternal during pregnancy)
- More in boys .
- Premature baby or VLBW baby .
- Cystic fibrosis.
- Congenital anomalies of the heart or lung.
- Immune deficiency is "rare".

RESPIRATORY INFECTIONS

- These are the **most frequent** infection of childhood
- Approximately 80% of all respiratory infections are URTIs (infants and primary school age children have as many as **10–12 per year**)
- **About** 80– 90% of childhood respiratory infections are **viral**.

Classification of respiratory infections:

A. Upper respiratory tract infection include (common cold "coryza", pharyngitis tonsillitis, acute otitis media, sinusitis "uncommon")
B. Lower respiratory tract infection (Bronchiolitis)
C. Infection of the lung parenchyma (Pneumonia)

1. The common cold (coryza):

- Is the commonest infection of childhood
- The commonest pathogens are viruses, **Rhinoviruses** (>100 serotypes) coronavirus – RSV.
- **C/P:** Clear or muco purulent nasal discharge , nasal blockage and low grade fever.
- **Rx:** Self limiting "No antibiotic", paracetamol,increases fluid intake.

NOTE

- (Cough may persist for up to 4 weeks after a common cold)
- No role of antitussive –vit C – decongestant in management

2. Pharyngitis (Tonsillitis or tonsillopharyngiti):

Causes:

- Viral (adenovirus – Rhinovirus – enterovirus EBV) common .
- Bacterial (group A beta haemolytic streptococci) rare < 3yrs.

Clinical picture:

- General – Fever (++ in bacterial), anorexia and headache.
- Local – Dysphagia, sore throat, enlarged tender lymph nodes and abdominal pain .

NB: It is not possible to distinguish clinically between viral and bacterial causes. (Centor criteria or score may help)

Treatment:

1. **Symptomatic Rx** – Antipyretic (paracetamol),increases fluid intake.
2. **Specific Rx** – Penicillin for 10 days in case of streptococcal infection
3. **Surgical** – **Tonsillectomy**

Indication :–

1. Peritonsillar abscess(Quinsy)
2. obstructive sleep apnea ,
3. frequent tonsillitis (→ 7 episodes/1yr , or → 5 episodes /every year for 2yrs , or →3 episodes /every year for 3yrs)
4. Tumor

Amoxicillin is best avoided as it may cause a widespread maculopapular rash if the tonsillitis is due to infectious mononucleosis. (Illustrated textbook 6th edition)

3. Acute otitis media

Definition: Is a suppurative infection of the middle ear cavity.

It's common [Most children will have at least one episode of acute otitis media and up to 20% will have three or more episodes]

Causes:

- Viral – RSV and Rhinovirus
- Bacterial – **Strep Pneumoniae** (40%), H. influenza (non typeable) and moraxella catarrhalis.

Epidemiology:

- Sex – more in boys.
- Age – 6 to 24 months (peak age).

Risk factors:

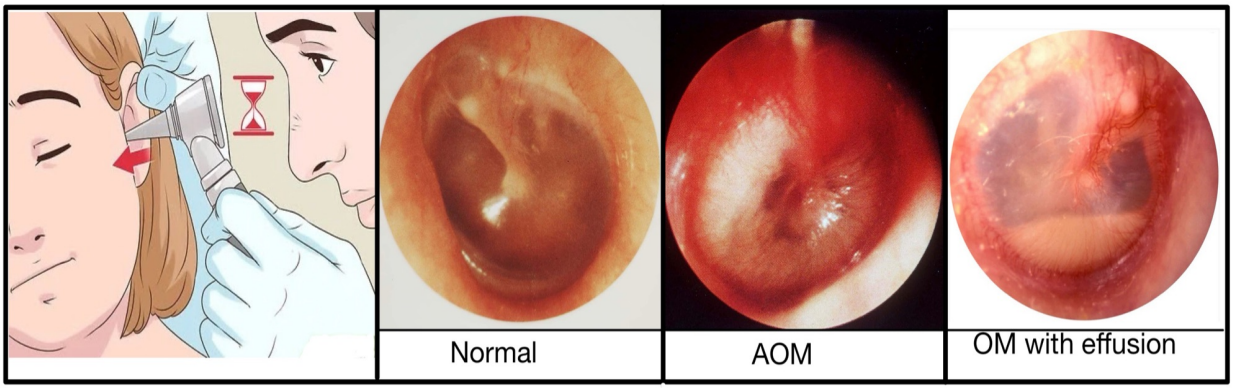
1. Infant, young children (B/C Eustachian tubes are short, horizontal & function poorly)
2. Bottle feeding (breastfeeding is protective)
3. Craniofacial anomalies as Cleft palate
4. Down syndrome
5. Exposure to tobacco smoking
6. Pacifier use ,immunodeficiency

Clinical presentations:

- **Infant**: Fever, irritability, poor feeding and vomiting,diarrhea , ear discharge.
- **Older children**: Fever, otalgia (ear pain) and otorrhea (ear discharge)

Diagnosis:

- History
- **Otoscopic examination** – congested & bulging of tympanic membrane with loss of the normal light reflection , sometimes perforation with discharge may be seen
- Pneumatic otoscopy T M movement



Note: every infant and young child with a fever —> should have their tympanic membranes examined to exclude AOM .

Treatment: Otitis media usually **resolves** spontaneously.

- **Antipyretics and analgesics** paracetamol or ibuprofen.
- **Antibiotics**: : **Amoxicillin** is DOC, (80–90 mg /kg /day), indication??

Antibiotics marginally **shorten** the duration of pain but have **not** been shown to reduce the risk of **hearing loss. Neither decongestants nor antihistamines are beneficial.**
Also, **No role of local medication* ear drops* in ttt of AOM.**

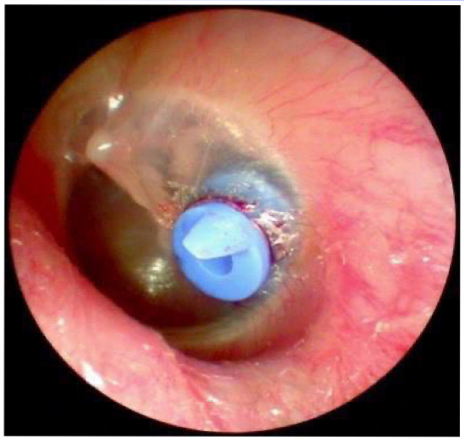
Complication of AOM:

- Recurrent otitis media with effusion (**Glue ear**)
- Conductive Hearing loss and speech and learning difficulties
- Acute mastoiditis
- Meningitis and dural vein thrombosis

Recurrent Otitis media with effusion (glue ear)

- Recurrent AOM leads otitis media with effusion (fluid level)
- Is the most common cause of **conductive** hearing loss in children
- It's common between the ages of 2 and 7 years. It usually resolves spon– taneously, but may cause conductive hearing loss and interfere with normal speech development
- **Treatment:** watchful waiting for 3 months, but if not improved ENT consultation & myringotomy with grommets tube insertion

Acute mastoiditis


myringotomy and grommet tube insertion

4. Sinusitis

Definition: Is a suppurative infection of the paranasal sinuses.

Cause:

- **Viral** as rhinovirus (usually preceded by URTI)
- **Bacterial** as strep pneumonia – Hib – staphylococcus aureus.

C/P:

- Fever, headache and cough.
- Muco purulent or watery discharge.
- pain, swelling and tenderness over the cheek from infection of the maxillary sinus.

Diagnosis	Treatment
• Plain X ray of paranasal sinus . • CT scan paranasal sinus. • Sinus aspiration C & S	• Analgesia. • Antibiotics.

Complication:

1. Orbital cellulitis.
2. Dural sinus thrombosis and meningitis.
3. Osteomyelitis of the frontal bone.
4. Exacerbation of Br asthma.

Key Points: **Pneumatization of the paranasal sinuses**

a. Maxillary and ethmoid sinuses – present at birth.
b. Sphenoid sinuses – Begins at 2 – 3 years and completes by 5 years.
c. Frontal sinuses – Begins to develop at 7 years of age and complete by age 12 years (frontal sinusitis is uncommon in the first decade of life).

STRIDOR

Two types of stridor are acute stridor and chronic stridor

Acute stridor:

(is an **emergency** the most important step in management is to maintain airways patent)

Causes:

1. Croup = Laryngio –tracheo–bronchitis .
2. Acute epiglottitis
3. Bacterial tracheitis
4. F.B.A in upper airways (Laryngeal)
5. Angioneurotic odema
6. Hypocalcemia
7. Corrosive (caustic) ingestion,Inhalation of smoke and hot fumes in fires
8. Infection (measles – Diphtheria – Glandular fever – retropharyngeal abscess– Quinsy)

mnemonic

COFFEE

- **C** → **C**roup –hypo **C**a–**C**orrosive ingestion.
- **O** → **O**dema (Laryngeal)
- **F** → **F**BA
- **F** → In**F**ection (bact tracheitis –measles – Diphtheria – Glandular fever – retropharyngeal abscess–Quinsy)
- **E** → **E**piglottitis .

Chronic stridor:

Causes:

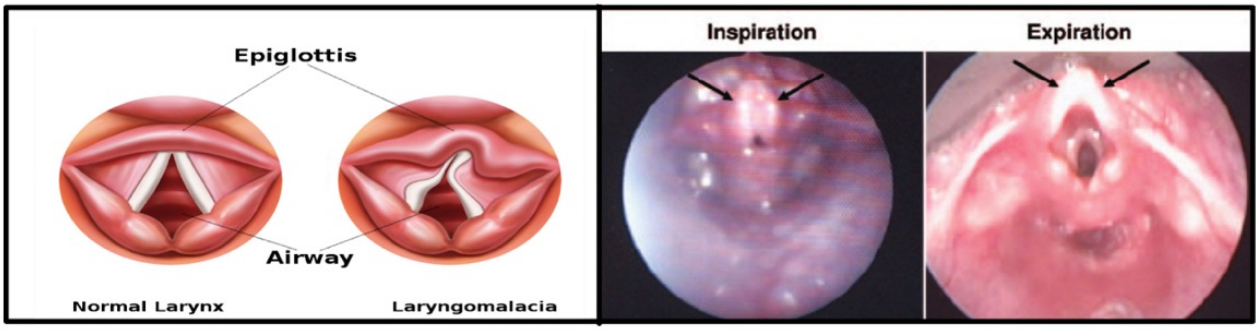
- Laryngomalacia
- Laryngeal web or hemangioma .
- Vocal cord palsy.
- Vascular ring.
- Subglottic stenosis.
- Compression from outside (goiter, lymphadenopathy and mediastinal mass)

mnemonic

- **2L** (Laryngomalacia –Laryngeal web)
- **2V** (Vocal cord palsy –Vascular ring)
- **Inside** (Subglottic stenosis–hemangioma)
- **Outside** (goiter, lymphadenopathy and mediastinal mass)

Laryngomalacia

- Is the **most** common cause of chronic and congenital stridor in infancy.
- Laryngomalacia means **soft Larynx** is a congenital disorder that leads to collapse during inspiration → stridor .
- Stridor {noisy breathing}started at birth or shortly after birth 4–6 weeks.
- Stridor usually (↑ with crying – URTI– Supine position) and (↓ in prone position) "SIDS?
- The infant usually is thriving well ,no other manifestation, O2 sat normal.



Diagnosis: (History + Examination) and flexible Laryngoscopy (omega like).

Treatment:

It's a **self limiting** condition improved spontaneously by the age of 18–24 months and **reassurance** of parents.

Laryngeal web

- It's congenital membrane covered by connective tissue between vocal cords
- Stridor present at birth

Treatment → Laryngoscopic ablation.

Subglottic stenosis

- Is the narrowing of the trachea below the vocal cord.
- It may be (**congenital** or **acquired** → cuffed ETT)

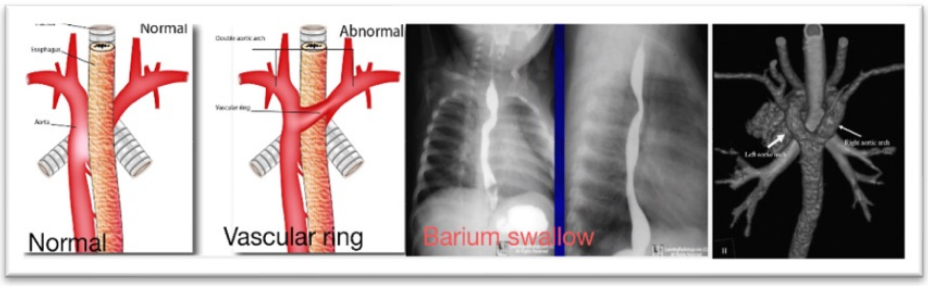
C/P → Stridor (insp + Exp)
Dx → –by bronchoscopy
Rx → mild (conservative) and sever (surgery or tracheostomy)

Vascular ring

Is congenital anomaly of the aortic arch and great vessels
(double aortic arch = 90% of cases)

C/P: arise from compression of the **trachea** & **esophagus**
→ Stridor – Apnea – Wheezing – Cyanosis – Dysphagia

Dx →Chest X ray, **barium swallow**, echocardiography and **angiography**.



Rx → Surgery

Viral CROUP

[Laryngo – Tracheo – Bronchitis]

- ✓ Croup is a common viral infection of the respiratory tract.
- ✓ Croup is the most common cause of Acute stridor in children.
- ✓ Viral croup accounts for 95% of Laryngotracheal infection.

Causes:

Parainfluenza virus (type 1–2–3) is the most common .
Other as RSV–Influenza virus–Metapneumovirus–Adenovirus

Epidemiology

Age: 6m – 6 years (peak 1– 2yrs)
Season: – common in autumn and early winter
Sex: slightly more common in males.

Pathophysiology:

Laryngotracheal airways inflammation – mucosal inflammation – Increase secretion (larynx, trachea → edema → narrowing → obstruction of the airways)

Clinical picture:

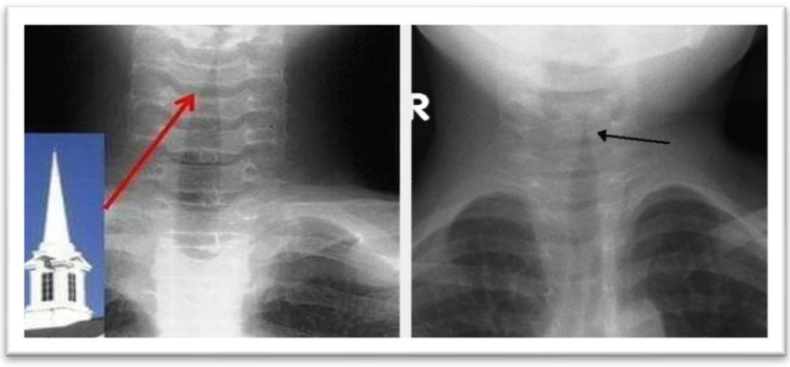
1. Gradual onset (started with symptoms of URTI (coryza) for 1–2 days "runny nose – low grade fever".
2. Followed by barking "brassy" cough, hoarseness of voice, harsh stridor (the symptoms often starting, and being worse, at night).
3. In severe cases – may patient develop (continuous stridor (insp + exp) – RD – cyanosis – restlessness}

Diagnosis: mainly **clinically** (history + examination).

N.B in any patient **suspected upper airways obstruction:**

- **Don't** examine the throat by tongue depressor or spatula.
- **Don't** put the patient in a flat position.
- **Don't** lose time in unnecessary blood investigation or radiological requests (priority is airways management)
- **Keep** patient calm as you can (mother's lap)

steeple sign in AP Neck view



Management:

Most cases of croup are treated at home (admission is often required in severe cases)

	Mild croup	moderate croup	severe croup
C/P	• cough mainly • No RD signs	• cough +stridor • Moderate signs of RD	• stridor at rest • Severe signs of RD (dusky, irritable, restlessness) • low O2sat
Treatment	1. at home 2. Inhalation of warm moist air(not beneficial) 3. Oral dexamethasone if needed **NB:** Parents should be advised to observe the child closely at home for signs of increasing severity.	1. humidified O2 2. **steroid:** reduce the severity and duration of croup a. **dexamethasone** Oral or injection [single dose] b. **Budesonide** nebulizer 3. Home observation	**as moderate +** 1. **Adrenaline nebulizer (racemic epinephrine)** Rapid but transient improvement due to short duration of action (so keep child in observation for 3–4hrs to ovoid rebound phenomenon) 2. **Intubation &MV** in very severe cases [very rare in viral croup]

The **decision to manage** the child with CROUP either at home or in hospital **depends on:**

1. Severity of the illness.
2. The time of day.
3. Ease of access to hospital.
4. The child's age (child < 12 months have narrow airway caliber → more needs admission)

Complication → viral pneumonia (2%)

Prognosis → excellent (recovery usually occurs in 2–5 Days)

Prevention → No vaccine for parainfluenza viruses

	Viral croup	Spasmodic croup
Cause	viral (parainfluenza)	allergy ? Atopy
Attacks	usually single but can be recurrent	recurrent
H/O„URTI coryza/ fever	**+ ve**	**– ve**
Treatment	at home steroids –adrenaline	same as viral croup

ACUTE EPIGLOTTITIS

Definition:

- ✓ Is acute inflammation of the epiglottis,
- ✓ now very rare since children in Libya are immunized against H. influenzae type–b (Hib)

Causes: an

- H.Influenzae type B (commonest cause)
- Streptococcus group A, C
- Staph aureus

Age: 2–7 years (but can all age groups)

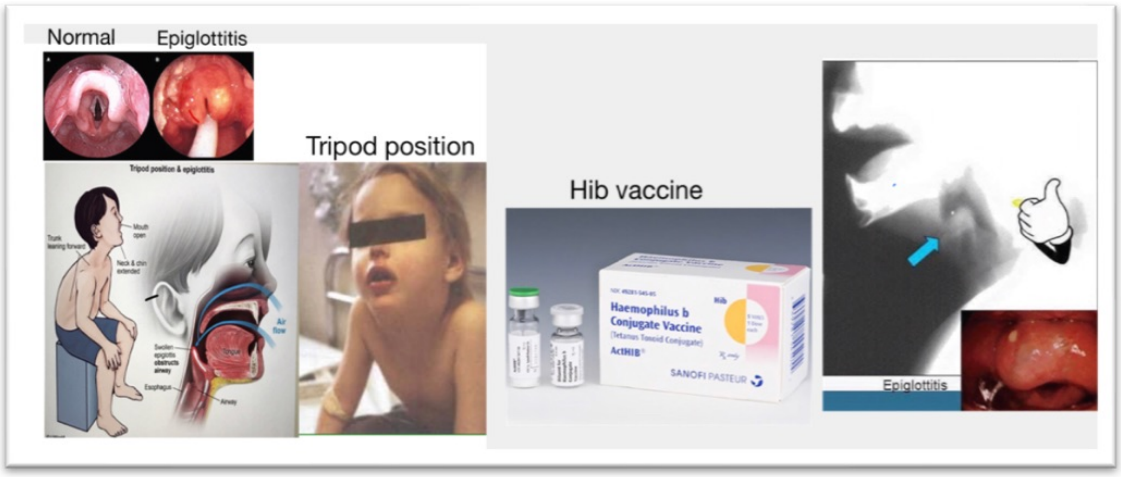
Q. Epiglottitis is a medical emergency – why?
A: Because of the risk of sudden airway obstruction)

Clinical picture: **Sudden** onset

1. Toxic child with high grade fever > 38.5 C°
2. Drooling of saliva (severe dysphagia)
3. The child severely exhausted
4. Muffled voice (hot potato voice) – stridor is mild & soft – No or little cough.
5. Tripod position (patient sitting with head held forward the mouth open and the jaw thrust forward)

Epiglottitis clinical picture :

3D (**D**istress – **D**rooling – **D**sphagia)
3T (**T**oxic – **T**emp high – **T**ripod position)



Diagnosis: If you suspect epiglottitis – Urgent admission to ICU

- CBC: Increase WBC mainly neutrophils
- Blood C & S → +ve for H influenza B
- Lateral neck X Ray "Thumb sign", not done.
- Direct observation under GA "Done only in operating room"

NB: Contraindicated – any attempts to visualize the epiglottis by examining the throat using tongue depressor – B/C it provokes airway spasm → total obstruction.

Management: It's medical emergency

- **A**: airways – **secure airways: ETT** (Rarely, this is impossible and urgent tracheostomy is required)
- **B**: Breathing – O_2 inhalation
- **C**: Circulation – IVF, **Antibiotic (3rd generation cephalosporin IV),** blood C&S .

Direct admission to ICU and call for (Anesthesia doctor – ENT doctor & pediatric consult) Under GA – Examine the airway with laryngoscope – cherry red color & swollen epiglottis.

N.B

- Steroids & adrenaline are ineffective in treatment of acute epiglottitis
- Introduction of Hib vaccine (2– 4 – 6 months) → Lead to decrease the incidence of acute epiglottitis worldwide.

Prevention:

1. Rifampicin prophylaxis to close contacts
2. Hib vaccine (2 – 4 – 6 months)

	Croup	Epiglottitis
Cause	Viral parainfluenza	Bacterial (HiB)
Onset	Days (Insidious)	Hours (abrupt)
Prodrome	Coryza	None
Appearance	Unwell	Toxic, very ill
Feeding	Can drink	No
Drolling of saliva	No	Yes
Cough	Barking	Absent or slight
Fever	< 38.5 C°	> 38.5 C °
Stridor	Harsh, Rasping	Soft, whispering
Voice, cry	Hoarse	Muffled (weak or silent)
Age	6m – 6 years	2 – 7 years
Blood C & S	– ve	+ ve
R_x	Symptomatic	ICU, Intubation, antibiotic

Bacterial tracheitis – Pseudomembranous croup (mcq)

- It's rare but serious infectious cause of upper airway obstruction
- **Age** – any age (3 – 10) years
- **Cause** – ***Staph aureus*** and strep pneumonia (may preceded by viral URTI)
- **C/P** – High grade fever, ill & toxic, stridor with mild cough (barking)
- **I_x** – CBC – CRP – Tracheal secretion C & S

ACUTE BRONCHIOLITIS

Definition:

- ✓ Is acute inflammation of the bronchioles.
- ✓ Is the commonest serious respiratory infection in infancy.

Etiology:

1. **Respiratory syncytial virus – RSV (80 %)**
   - **RSV**
     - ✓ TYPES (A &B)
     - ✓ Highly contagious
     - ✓ IP :2–6 days
     - ✓ No vaccine available
2. Other (rhinovirus, adenovirus, influenza virus, and human metapneumovirus, Mycoplasma pneumonia)
3. ***co-infection*** with more than one virus, particularly RSV and human metapneumovirus, may lead to a more **severe** illness.

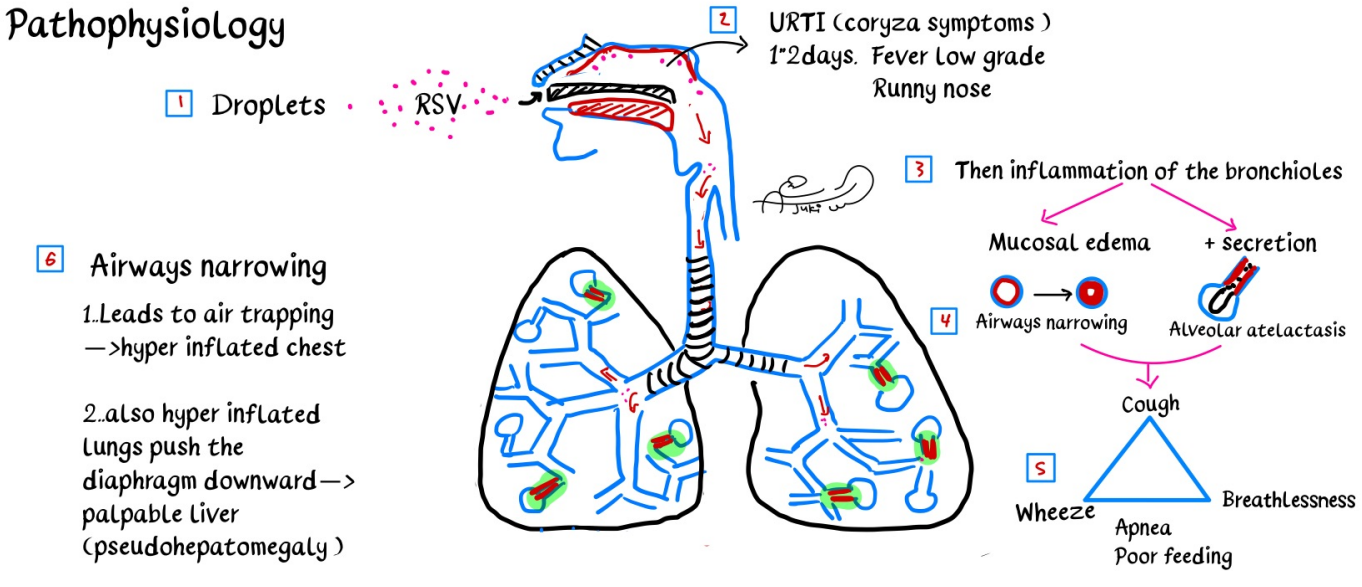
Epidemiology:

- Age → < 2 years
- 90% are aged 1–9 months (peak age 3–6 m)
- Season: winter and early spring
- Sex: slightly more in ♂ (1.25 ♂ : 1 ♀)

Pathogenesis:

Invasion of the mucosa & submucosa by virus → inflammation of bronchioles → mucosal edema & accumulation of exudates → Narrowing of the diameter of bronchioles – Obstruction (Bilateral) – airtraping & overinflation.

Pathophysiology



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Airways narrowing	+ secretion
Ronchi with prolonged expiration	Fine end inspiratory crepitation or crakeles

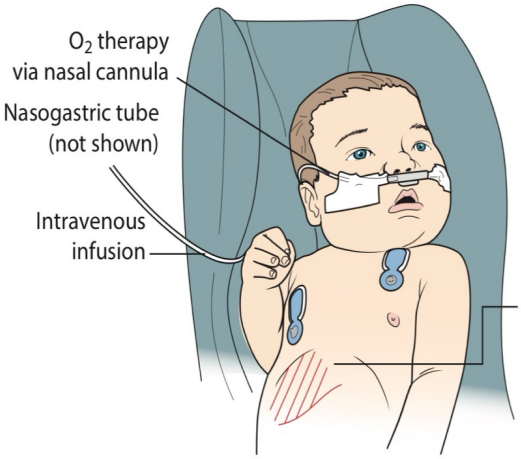
Clinical pictures:

Symptoms:

1. URTI (coryzal symptoms) for few days
2. Followed by cough – Wheeze – Breathlessness.
3. Decrease feeding & Irritability
4. Apnea may be more prominent early in young infant

Signs:

1. General Examination → Signs of RD + Irritable, may be dehydrated? & Audible wheeze
2. Chest Examination – tachypnea and recession SCR– ICR – SSR.
3. Chest findings on auscultation – Decreased air entry bilateral, vesicular breathing with prolonged expiration, bilateral Rhonchi and fine crackles (crepitation) may be heard.
4. The liver may be palpated as it is displaced downwards due to hyperinflation.



Clinical features

Dry, wheezy cough

Cyanosis or pallor

Tachypnoea and tachycardia

Subcostal and intercostal recession

Hyperinflation of the chest:
- sternum prominent
- liver displaced downwards

Auscultation:
- fine end-inspiratory crackles (crepitations)
- prolonged expiration/ wheeze

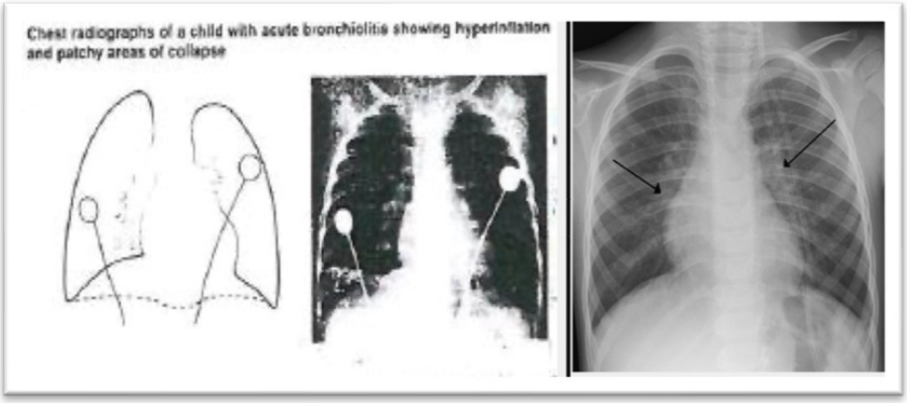
Clinical features of bronchiolitis in an infant requiring hospital care.

Causes of acute respiratory distress in an infant:
- **Bronchiolitis**
- **Viral episodic wheeze**
- **Pneumonia**
- **Heart failure**
- **Foreign body**
- **Anaphylaxis**
- **Pneumothorax or pleural effusion**
- **Metabolic acidosis**
- **Severe anaemia.**

Diagnosis: the diagnosis Mainly **clinically** (based on history & clinical finding)

Usually No investigations are routinely recommended. In particular, chest X-ray or capillary blood gasses are only indicated if respiratory failure is suspected.

1. **Chest X Ray:** may be Normal or "Hyperinflated Chest", or atelectasis bilaterally. Chest radiographs of a child with acute bronchiolitis showing hyperinflation and patchy areas of collapse



2. **CBC** – Increase WBC mainly Lymphocytes or Normal WBC
3. **Immunofluorescence of nasopharyngeal swab → PCR** – for detection of virus (not routine) B/C Little effect on management
4. **ABG** – in severe cases

Management–

treatment is according to severity (If mild → Rx at home, If sever → Admission)

Indication of Admission

1. Severe respiratory distress (RR > 60–70 c/min), grunting, cyanosis.
2. apnoea (observed or reported)
3. Poor feeding infant, inadequate oral fluid intake or dehydrated.
4. persistent oxygen saturation of <92% in room air.
5. Infant < 3–6 months of age.
6. High risk infant:
    a. Preterm & VLBW < 1.5 Kg
    b. Immune deficiency.
    c. Congenital heart disease.
    d. Chronic Lung disease, BPD

Treatment of acute bronchiolitis is supportive

1. Good hydration IVF or NGT feeding.
2. Good oxygenation (humidified O_2) is delivered via nasal cannulae or head box at a concentration adjusted according to pulse oximetry.
3. Close observation (sign of RD – RR–apnea – Pulse oximetry)
4. In some infants, non–invasive respiratory support with continuous positive airway pressure (CPAP), bilevel positive airway pressure (BiPAP) or mechanical ventilation is required in ICU .

Specific R_x:

Ribavirin nebulizer (Antiviral) "Not routinely used".
Indicated only in : High risk infants.

N.B:

- **Bronchodilator** (salbutamol or ipratropium) are effective in many studies.
- Corticosteroid and Antihistamine have no benefit.
- Antibiotics has no role (except if 2ry bacterial infection).
- Physiotherapy has no role.
- Hypertonic saline nebulizer has no role!

Complication:

1. Apnea and respiratory failure
2. Dehydration
3. 2ry bacterial infection
4. Pneumothorax
5. Recurrence
6. Bronchiolitis obliterans (adenovirus 3 & 7)

Prevention:

1. Hand hygiene and avoid exposure to symptomatic patient (as RSV is highly infectious)
2. RSV immunoglobulin or monoclonal antibody (palivizumab) given monthly IM to children < 2yr for high risk group especially preterm babies.
3. No vaccine available against RSV.

Prognosis:

a. Most cases recover within 1–2 weeks.
b. Mortality ≈1 % due to RF, Apnea and dehydration.
c. Reinfection with RSV may occur.

PNEUMONIA

Definition: Inflammation of the lung parenchyma.

It is a major cause of childhood mortality in low- and middle-income countries.

Classifications:

A. According to Etiology(causative organisms):-

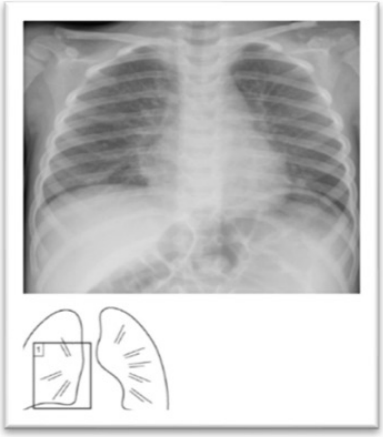
1. Viral	2. Bacterial	3. Other
• RSV • Influenza • Adenovirus,measles • COVID-19, VZV.	• Strep Pneumonia • Staph aureus • H. Influenza • Mycoplasma Pneumonia • Chlamydia pneumonia	• TB pneumonia. • Fungal – aspergillosis • Protozoal – as pneumocystic carinii in HIV patients.

Clinical findings in viral versus bacterial pneumonia

	Viral	Bacterial
Temperature	↑	↑↑↑
H/O URTI	+ +	–
Toxicity	+	+ +
WBC	N or ↑ mainly lymphocyte	↑↑ Mainly neutrophile
Diagnosis X-Ray	Interstitial	Lobar

.N.B (In clinical practice, it is **difficult to distinguish between viral and bacterial pneumonia**)

B. According to Anatomical or Radiological appearance:-

Lobar pneumonia	Bronchopneumonia	Interstitial pneumonia
• Homogenous opacity in one or more lobes. • Usually bacterial	• Scattered opacities in both lungs. • Can be viral or bacterial.	• Scattered Bilateral perihilar pulmonary infiltrate. • Usually viral.
		

C. According to age:

Newborn	Infant and young child	Child > 5 years
(Bacterial **organisms** from the mother's genital tract)	**Viruses** are the most common	**Bacteria** are the most common
• Group B streptococci • G –ve enterococci (E coli)	• **Viral** – RSV • Bacterial – Strep pneumonia*, H. Influenza and staph aureus, Bordetella pertussis and Chlamydia trachomatis	• **Bacterial** – Mycoplasma pneumonia, strep pneumoniae and chlamydia pneumoniae • Viral – RSV

N.B (At **all ages Mycobacterium tuberculosis TB** should be **considered)**

D. According to source of infection

- Community acquired pneumonia
- Hospital acquired pneumonia

Important types of bacterial pneumonia

Strep pneumonia	staphylococcus pneumonia	chlamydia trachomatis	mycoplasma pneumonia
*Age of onset: (all age groups) *C/P typical pneumonia, cough. fever, breathlessness *Chest X ray: typical lobar pneumonia *R_x – supportive + antibiotic (Amoxicillin or co-amoxiclav)	*Peak age: (infancy < 1 year) *C/P sever (sudden onset & rapid deterioration with high grade fever & bad general condition *Complication – lung abscess – pleural effusion empyema – pneumatocele *R_x – supportive + antibiotic (cloxacillin or vancomycin)	*Age: (1st 3 months of age) *Gradual onset – No fever + conjunctivitis (**sticky eyes**) Staccato cough *CBC: Eosinophilia + serology *R_x – supportive + antibiotic (macrolides)	*Age: 5 to 15 years (school age) *Insidious onset with cough & low-grade fever ***Extra pulmonary manifestation:** • **Skin:** Erythema multiforme – Steven Johnson syndrome • **CNS:** Meningoencephalitis – gallian barre syndrome • **Joints:** Arthritis • **Heart:** Myocarditis – pericarditis • **Blood**: Autoimmune hemolytic anemia **GIT** – Protein losing enteropathy

Risk factors:

1. Anatomical lung problems e.g H type TEF
2. Chronic lung disease e.g cystic fibrosis
3. Immunodeficiency
4. Neuromuscular disorders as CP ,myopathy
5. GERD
6. Congenital heart disease[Left to Right shunt –>lung congestion]
7. Contaminated equipment e.g intubation

Diagnosis:

1. **Chest X Ray** – Radiological **worse** than clinical (**walking pneumonia)**
2. Cold agglutinins

Rx– **Macrolides**

Clinical Pictures

Symptoms: (Fever, cough and breathlessness are the most common presenting symptoms)

A. **General:** Fever – malaise – poor feeding – vomiting & localized chest, abdominal, or neck pain is a feature of pleural irritation.
B. **Pulmonary:** Cough (Dry → then productive), Breathlessness and Grunting

Signs:

A. **General:** **Tachypnea** & Sign of RD – unwell child – sometimes dehydrated
B. **Chest:**

1. **By inspection** → decrease movement on the affected side.
2. **By palpation** → Trachea central, decrease chest expansion, increase TVF.
3. **By percussion** → Dullness (stony dullness if complicated by pleural effusion)

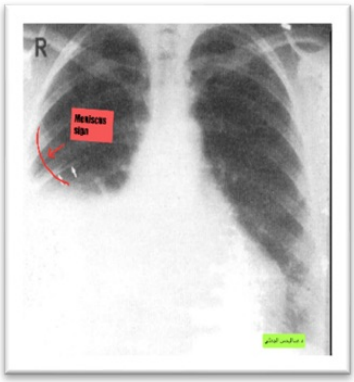
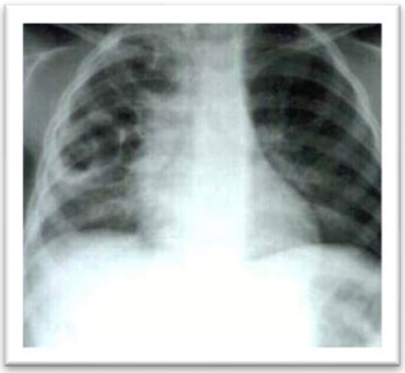
4. **By auscultation** → decreased air entry, bronchial breathing, cripitation (course type and increase vocal resonance)

- ✓ The **most sensitive clinical sign** of pneumonia is **raised respiratory rate (Tachypnea)** , so this must always be measured in a febrile child.
- ✓ The **classic signs of consolidation** are localized dullness on percussion, decreased breath sounds, bronchial breathing and end–inspiratory coarse crackles are often absent in **young children**,

Important types of pneumonia:

Diagnosis:

1. CBC – WBC high:
   - mainly neutrophils increased – Bacterial
   - mainly lymphocyte increased– viral, TB, pertussis
2. CRP titer: high
3. Chest X Ray: for diagnosis & identify any complications as lung abscess .

upper Rt lobar pneumonia	Bronchopneumonia	interstitial pneumonia
Rt side pleural effusion  Meniscus sign	Rt side lung abscess	pneumatocele (staph) 

Chest X-ray:

1. Necessary if there is doubt about the diagnosis.
2. Repeat X ray after 4–6 weeks to assess the resolution.
   - Neither a chest X-ray nor blood tests, including CBC & CRP, are able to reliably differentiate between a viral and bacterial cause.
4. CT scan chest may be needed

4. Investigations for causative organisms

   - **Viral** – Isolation of virus or detection of antigens in respiratory secretions by ELISA or PCR .
   - **Bacterial** – Isolation of organism from blood (Blood C & S)
   - **Cold agglutinins** – IgM titers for (mycoplasma pneumonia)

5. ABG – only in severe RD.

Treatment: Most of cases can be managed at home

Indication of Admission

- Severe RD –very high RR – grunting.
- Interfere with feeding – dehydration.
- Immunodeficiency.
- O_2 saturation < 92%.
- Recurrent Apnea.
- Family unable to provide good care at home or very young age < 3– 6 months
- Associated with complications such as lung abscess.
- No response to oral R_x

Supportive

- Humidified O_2 for hypoxia
- IVF if dehydrated or high RR.
- Antipyretic for fever
- Monitoring (RR –RD signs –apnea – pulse oximetry)
- Physiotherapy has no proven role (except in CF or bronchiectasis child)

Specific

Antibiotic – Choice of antibiotic depends on [**child's age – severity – C X Ray**]

- Newborn (IV) – Ampicillin + gentamicin / or 3rd generation cephalosporin.
- Older infant & young children – Amoxicillin or co-amoxiclav. (oral in mild &mod pneumonia)
- More than 5 years – either amoxicillin or/and macrolides (erythromycin, clarithromycin or Azithromycin)

Note: In case of pleural effusions (sterile parapneumonic effusion) usually resolve with appropriate antibiotics, but small cases develop an empyema require drainage with chest tube

Complication of pneumonia

1. pleural effusion.
2. lung abscess.
3. empyema.
4. pneumatocele
5. pneumothorax & bronchopleural fistula.
6. respiratory failure
7. meningitis–sepsis –septic arthritis –pericarditis.

Recurrent pneumonia:

defined as ≥2 episodes in one year or ≥3 episodes ever, with radiographic clearing of densities between episodes.

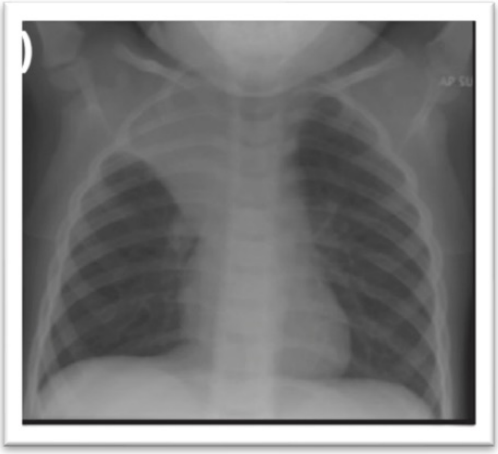
Causes:

A. On the same site: missed F.B.A (Foreign body aspiration)
B. On different sites:

- Immunodeficiency, malnutrition
- Cystic fibrosis, immotile cilia syndrome.
- Bronchiectasis.
- CHD and VSD.
- H type TEF.
- GERD.
- Hypotonia, Down syndrome and cerebral palsy.

Aspiration pneumonia

- Aspiration – is inhalation of either oropharyngeal or gastric contents into the lower airways
- Chemical pneumonitis – Is inflammation of lung due to chemicals as gastric content
- Aspiration pneumonia – Usually occurs in the **upper Rt lobe** and chest X Ray shows Rt apical opacity



Risk factors for aspiration

- GERD.
- Cleft palate.
- Neuromuscular disorders
  as myopathy or Cerebral Palsy " CP

  Down syndrome
- H type TE**F**

- In Libya History of **oil ingestion** in infant suspected aspiration pneumonia is important B/C some family using it for treatment their infant with coryzal symptoms (is very serious)

Rx of chemical pneumonitis:

- **Supportive R_x** – good hydration and good oxygenation
- **Antibiotic only** if 2ry bacterial infections

Note: child presented with H/O kerosene ingestion – Gastric Lavage is contraindicated.

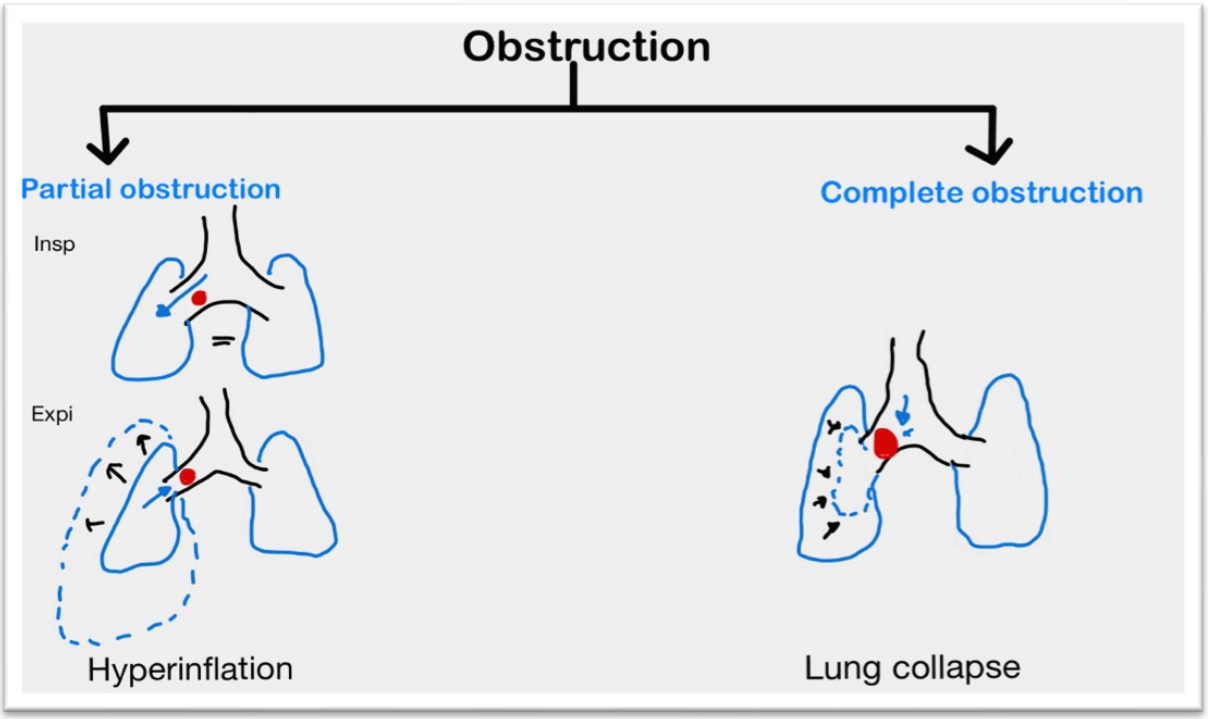
WHO Classification of acute respiratory illness (ARI)

1–Normal RR, No Lower chest indrawing	**No pneumonia (common cold or cough)**	Home care and no antibiotics
2–Increase RR – No Lower chest indrawing	**pneumonia**	Home care and antibiotics
3–Increase RR + Lower chest indrawing	**severe pneumonia**	Refer to hospital and antibiotics
4–Increase RR + Lower chest indrawing + **danger signs (cyanosis – convulsion – abnormal sleep + poor feeding–severe malnutrition)**	**Very severe pneumonia**	Refer urgently to hospital supportive therapy and antibiotics

Foreign body aspiration

- Common age – 6 months – 4 years (**peak age 1–2 yrs**)
- Common site – **Rt main bronchus** & Larynx
- Most common foreign body is **peanuts,** but can be popcorn small toys, or coins.
- Foreign body can be – **radiopaque** or **radiolucent** and irritant or non–irritant

Pathophysiology



Clinical pictures:

Symptoms:

- **Sudden** episodes of coughing and shocking & wheezing and breathlessness.
- If the FB in the upper airway – acute **stridor.**
- If **not diagnosed** (missed) from start – later on the child may be presented with persistent or recurrent cough, wheezing and recurrent pneumonia.
- When acute aspiration causes total or near total occlusion of the airway → lead to death or hypoxic brain damage.

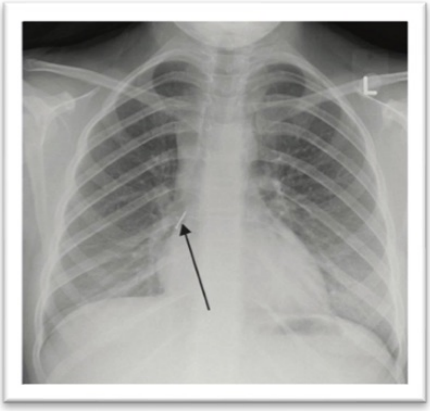
Signs: RD signs + **localized** or **unilateral** decrease entry or absent bronchi (it may be bilateral)

Diagnosis:

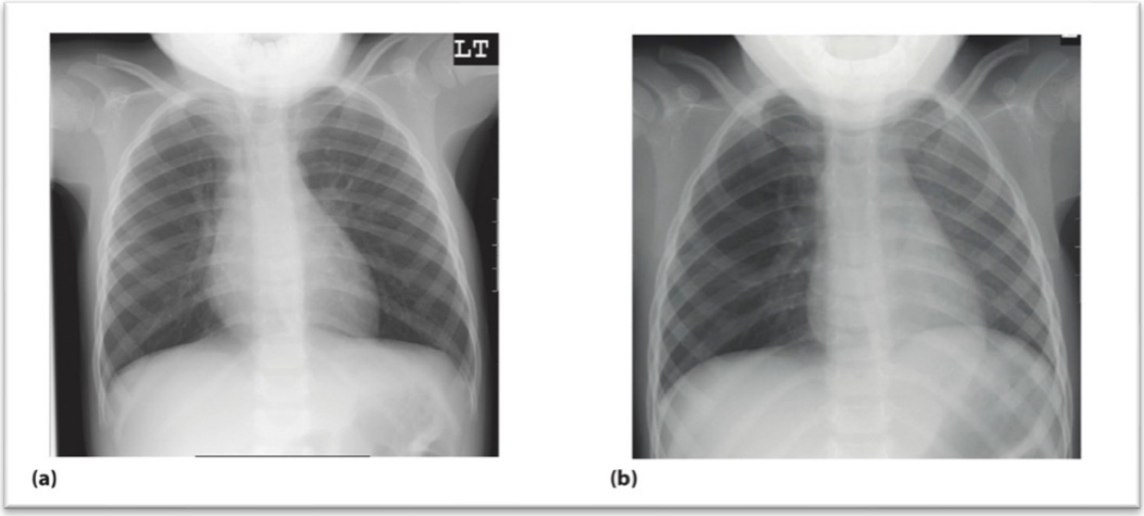
1. **Chest X Ray** (inspiratory & expiratory view)

- If partial obstruction → leads to air trapping (acting as a **ball valve)** → Hyperinflation on the same side

- If complete obstruction – lead collapse on the same side pulling trachea and mediastinum also.
- CT scan chest may need



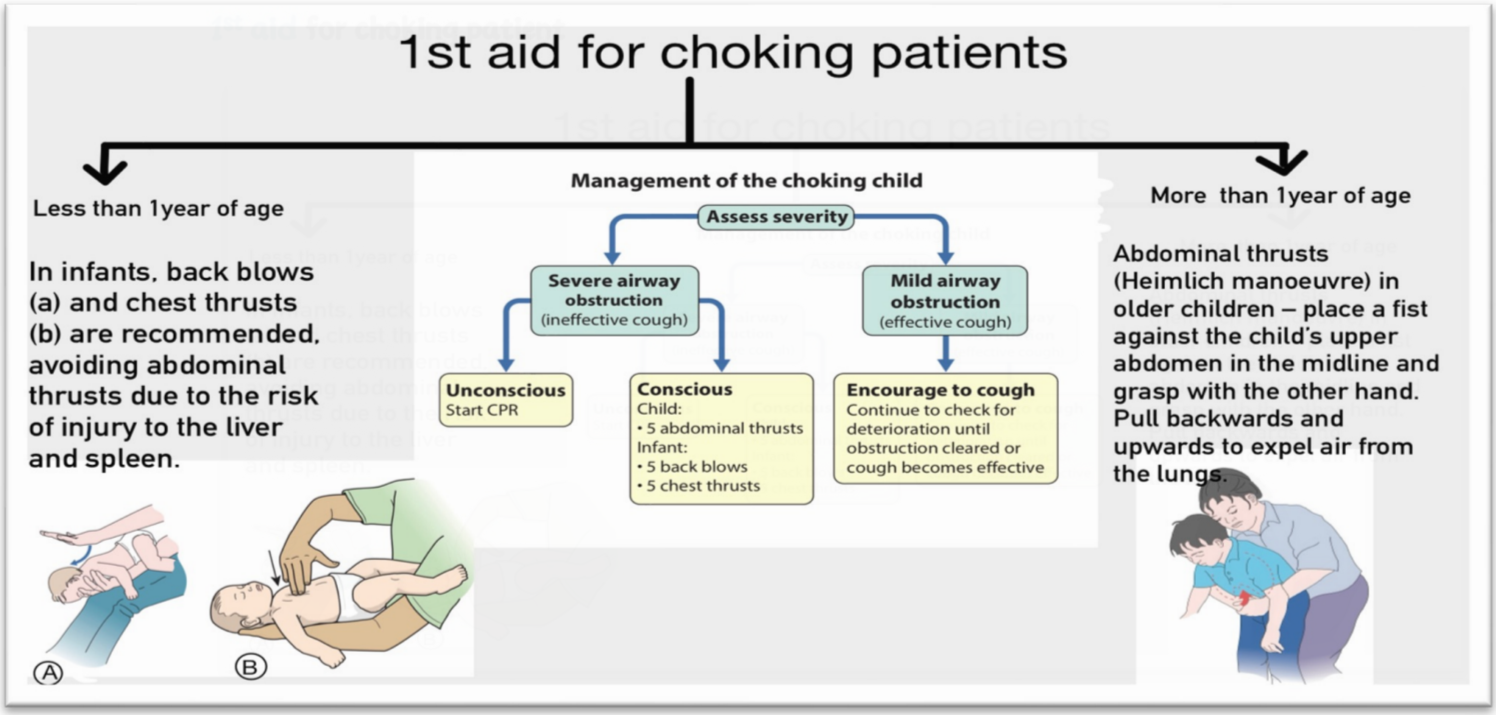
Chest X–ray showing a pin (arrow) in the right main bronchus



Peanut inhalation, with peanut in right main bronchus acting as a **ball valve**, so **inspiratory** film(a) is normal whereas **expiratory** film (b) shows hyperlucency of right lung and mediastinal shift.

2. Rigid bronchoscopy– (Diagnostic & therapeutic)

Management:



Definitive treatment of any FB is removal of this FB by bronchoscopy (ENT)

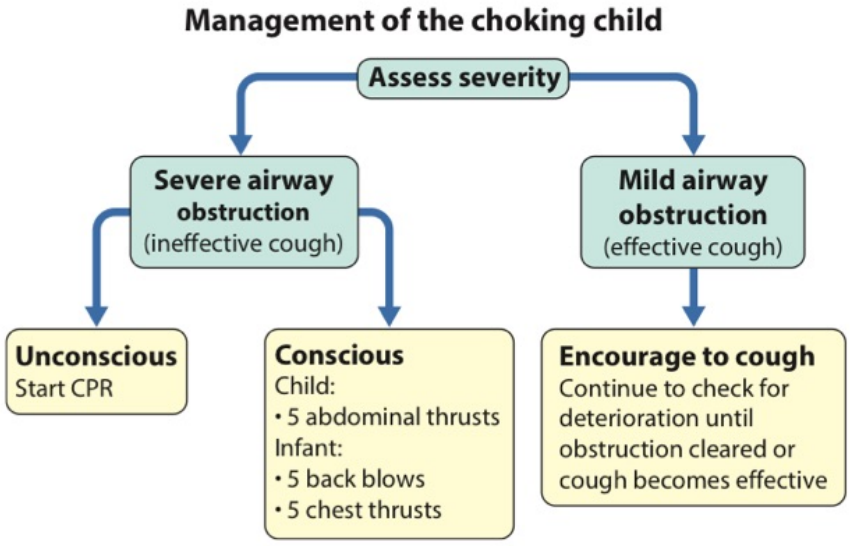
2 Types of bronchoscopies:

1–Rigid (diagnostic + therapeutic),

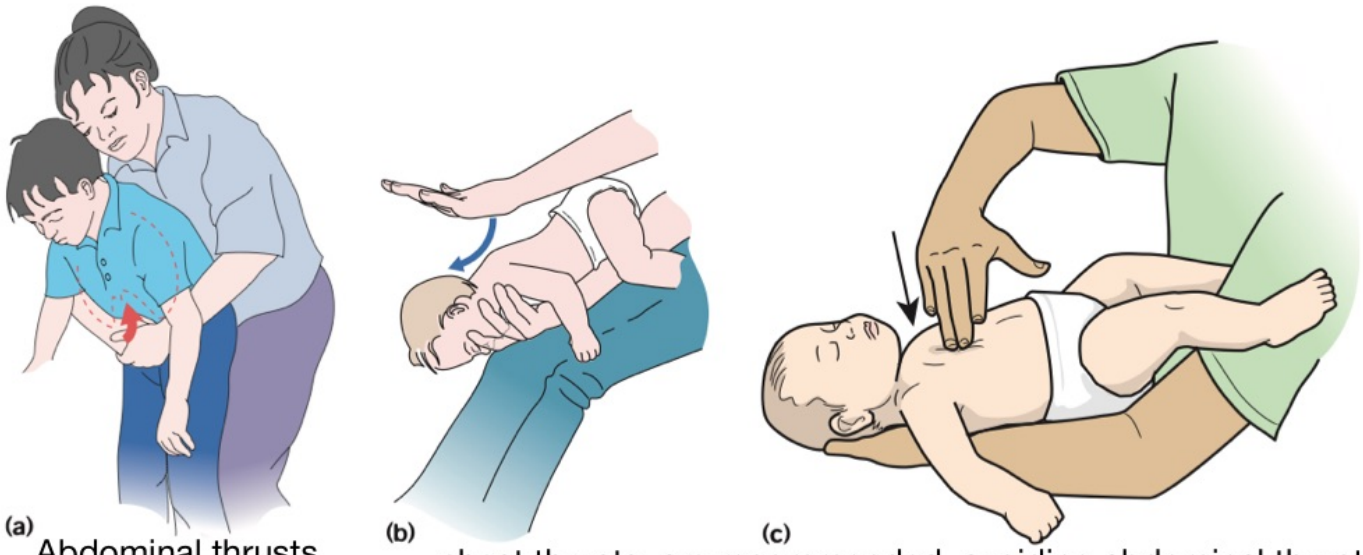
2–flexible (diagnostic mainly)

Complication:

- Complete airway obstruction – death
- Recurrent pneumonia
- Bronchiectasis(Localized)
- Lung abscess



Management of the choking child from a foreign body airway obstruction (FBAO).



Abdominal thrusts (Heimlich manoeuvre) chest thrusts are recommended, avoiding abdominal thrusts due to the risk of injury to the liver and spleen.

Bronchial Asthma

Definition: Chronic inflammatory airway disease characterized by hyperresponsiveness of airways to various stimuli which lead to reversible airway obstruction either spontaneously or by medication.

Why is bronchial asthma an important disease and should be controlled?

B/C 1–is an important cause of school absence.
2–it restricts child daily activity.
3–it causes anxiety for the child and family.

Epidemiology:

Incidence: 15–20% of children in UK (Asthma is the most common chronic respiratory disease in childhood)

Age: usually > 2 years (but can occur in any age)

prevalence data for asthma

- Male > female (pre–pubertal)
- Female > male (post–pubertal)
- Urban > rural
- Developed (western) > undeveloped countries

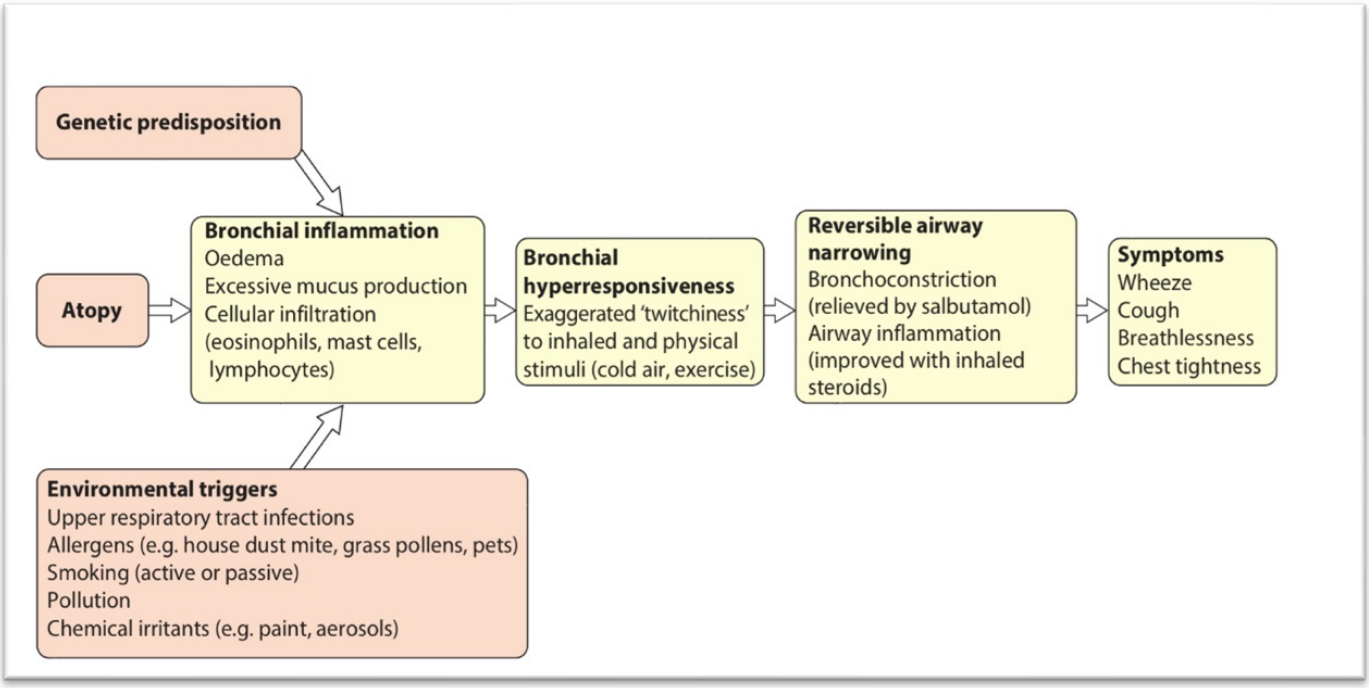
Risk factors:

- Genetic predisposition
- Non breastfeeding
- + ve F/H of atopy (allergic rhinitis – allergic conjunctivitis– **Eczema** – hay fever – food & drugs allergy.)
- Urban life – black race
- Parenteral smoking

Note:

1. ≃ 40% of children and infant who have viral wheezy infection – will have asthma through childhood
2. ≃ 80% of childhood asthma are associated with prior viral exposure

Pathophysiology :



Clinical picture

Symptoms

- Recurrent attacks /episodes of cough – wheeze and breathlessness (in the presence of the triggers factors) ± chest tightness.
- Patients are asymptomatic between attacks (interval symptoms)
- Usually symptoms increase at night (early morning), (cortisone circadian rhythm –dust mites)
- May be coughing is the only symptom called cough variant asthma.

Environmental triggers:- مهمة في الهيستوري

- Viral URTI (cold or flu).
- Allergens (dust mites –pollens –molds –pets …)
- Exercise (running, playing hard……especially in cold weather)
- Smoking (passive or active)
- Psychological factors (stress, emotions/ crying & laughing hard)
- Drugs (beta blocker, aspirin)
- Strong smell and chemical spray e.g. perfumes, paints.
- Cold air.

Examination: can be completely **normal between attacks**

General: Child tachypnic, restlessness and may be irritable and have sign of RD ± Audible wheeze

Chest:

1. Inspection:

- Barrel shaped chest
- Harrison sulcus (Horizontal grasive at insertion of diaphragm)

2. Palpation:

- Chest expansion – Decreased bilateral
- Tactile vocal fremitus (TVF) – Decreased

3. Percussion: Resonant note bilateral

4. Auscultation:

- Air entry – Decreased bilateral
- Type of breathing – Harsh vesicular breathing with prolonged expiration
- Added sounds – Rhonchi bilateral

Diagnosis: (mainly clinically especially in children less than 5yrs of age)

1. History – Compatible history with Br asthma

- Recurrent attacks
- + ve F/H of atopy
- Age usually > 2 years
- Improved by bronchodilator
- + triggers factors
- Nocturnal symptoms

2. Examination – as mentioned before

3. Investigation – support your diagnosis

- CBC – Eosinophilia may be present
- Chest X Ray – may be normal or shows hyperinflated chest.

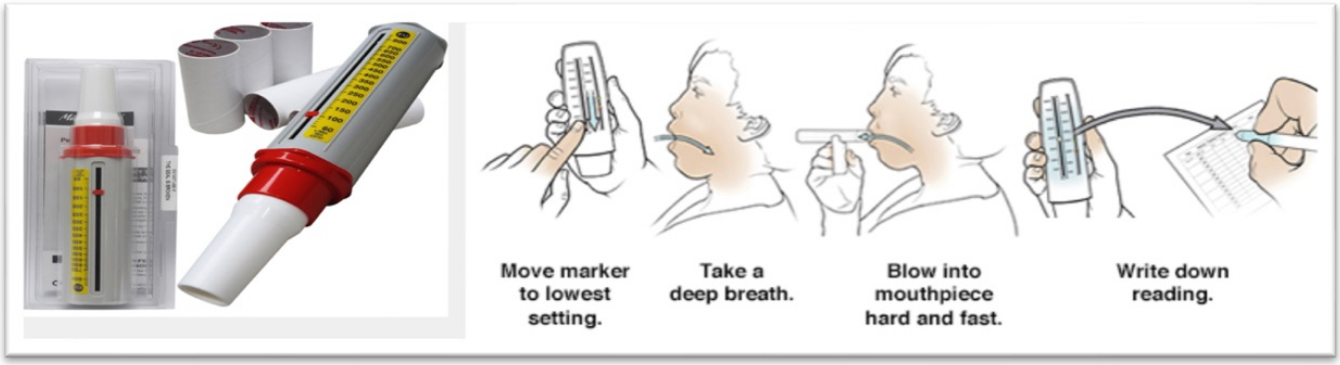
> **Note**: Chest X Ray is done only if you suspected:
>
> 1. Pneumonia
> 2. Sudden deterioration
> 3. Pneumothorax/Lung collapse.
> 4. before and after mechanical ventilation in PICU

- IgE level – high (normal level doesn't exclude)
- ABG → **Indication** (rapid Clinical deterioration, inability to maintain O2 sat >94 %, acute severe asthma – life threatening asthma)

(**ABG** – Early (Hypocapnia) CO2 washout B/C Hyperventilation & Late – Hypoxia + Hypercapnia+ acidosis)

- **Skin prick test** – By testing for common allergens – By injecting it intradermally – Look for induration
  - Diagnosis of atopy(allergy)
  - To identify allergen which may act as trigger
  - Help for immunotherapy

- **Peak expiratory flow meter PEFM** (used in children >5yrs used for diagnosis of asthma & monitoring response to treatment) FEV1 <80%, & >20% diurnal variation.
  which is simple to use, portable and useful for serial measurement



- **Pulmonary function test (spirometry) is** possible in children aged >5yrs & will shows Airflow limitation (obstructive airways disease)
  1. FEV1 (relative to percentage of predicted norms) FEV1/FVC ratio <80%.
  2. Bronchodilator response (to inhaled –agonist): Improvement in FEV1 ≥12%
  3. Exercise challenge: Worsening in FEV1 ≥15%

Management of bronchial asthma

Aims of asthma management:

- Control symptoms
- Prevention of exacerbation
- Normal activity of child e.g (School performance)
- PEFR near normal
- Prevention of irreversible air follow remodeling and limitation.
- Prevention of Br asthma mortality

Advice that should be given to asthmatic patient or parents

- Explain to the family (parents) the natural history of disease
- Avoid the triggers factors.
- Importance of regular follow–up
- When drugs should be used (regularly or 'as required').

- How to use the drug (inhaler technique) / (doses &duration) .
- What each drug does (relief vs prevention).

Classification of asthma according to severity

Mild	Moderate	Sever	Life threatening
Mild symptoms (wheeze)	Moderate symptoms	Severe Cough, wheeze and Dyspnea	Sever wheeze dyspnea
Mild RD	Moderate RD	Sever RD	Sever RD + Cyanosis
Mild decrease air entry, few ronchi	Moderate decrease air entry and expiratory Ronchi	*too Breathlessness to speak or feed *tachypnea *tachycardia *Pulsus paradoxus	*Drowsy *Agitation, confusion *Exhaustion *Silent chest *Cyanosis *Arrythmia, Low BP.
O2 sat > 92%	> 92%	< 92%	< 92%
PEFR > 80%	50 – 80%	33– 50%	< 33%
Rx home	Hospital	PICU	PICU

Management of acute Asthma exacerbation

management of acute asthma exacerbation

1-Good assessment
2-oxygen by mask high flow
3-SABA salbutamol nebulizer (dose 2,5mg <8yrs -5mg >5yrs)——15-20minutes ——>if no improvement (repeat salbutamol neb) ——15-20minutes ——>if no improvement (repeat salbutamol neb) add ipratropium bromide neb & start steroid [either prednisone oral 2mg /kg or hydrocortisone injection]

Assess response to treatment

Responding

- Continue bronchodilators (Ventolin inhaler)
- Continue oral preamsolone or 3-5 days

At discharge:
- Review medication and inhaler technique
- Provide personalized asthma action plan
- Arrange appropriate follow-up

In ttt of mild moderate asthma at home
1-Keep calm and reassure child and parents
2-Short-acting 2-agonist via spacer (with face mask for those under 3), 2–4 puffs, increasing by 2 puffs every 2 min to 10 puffs if required .+- prednisolone &assess response >15-30min.

Not responding:

Transfer to PICU

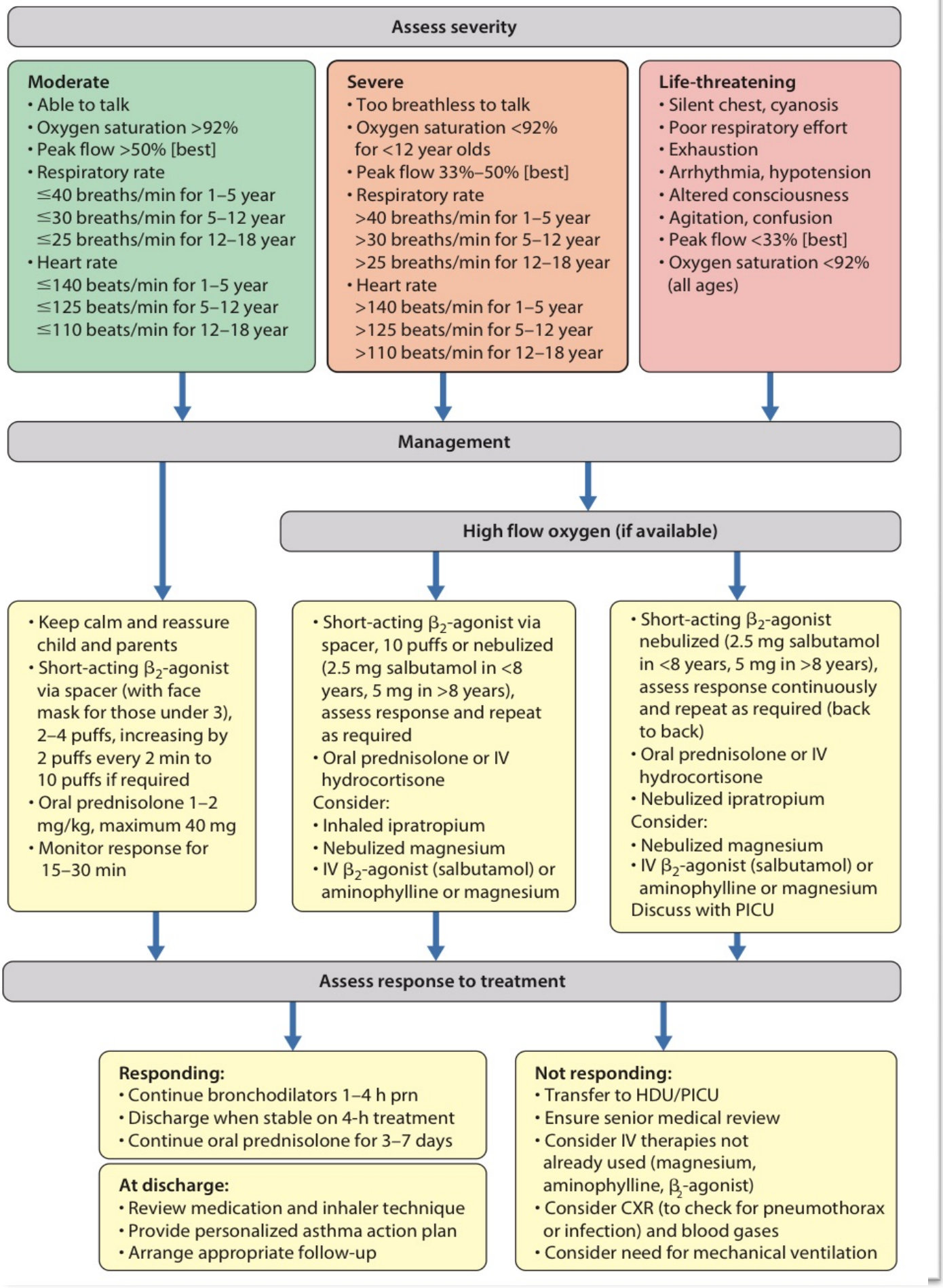
1-Connect pt to monitoring O2sat -ECG -Bp HR

2-continue neb medication back to back as required

3-start iv therapies not already used
*mg sulphate IV infusion slowly
*salbutamol IV infusion slowly
*aminophylline IV

4- do CXR (to check for pneumothorax ,collapse or infection) and blood gases ABG

5-Consider need for ETT & mechanical ventilation


Assess severity
Moderate
• Able to talk
• Oxygen saturation >92%
• Peak flow >50% [best]
• Respiratory rate
≤40 breaths/min for 1–5 year
≤30 breaths/min for 5–12 year
≤25 breaths/min for 12–18 year
• Heart rate
≤140 beats/min for 1–5 year
≤125 beats/min for 5–12 year
≤110 beats/min for 12–18 year
Severe
• Too breathless to talk
• Oxygen saturation <92% for <12 year olds
• Peak flow 33%–50% [best]
• Respiratory rate
>40 breaths/min for 1–5 year
>30 breaths/min for 5–12 year
>25 breaths/min for 12–18 year
• Heart rate
>140 beats/min for 1–5 year
>125 beats/min for 5–12 year
>110 beats/min for 12–18 year
Life-threatening
• Silent chest, cyanosis
• Poor respiratory effort
• Exhaustion
• Arrhythmia, hypotension
• Altered consciousness
• Agitation, confusion
• Peak flow <33% [best]
• Oxygen saturation <92% (all ages)
Management
High flow oxygen (if available)
• Keep calm and reassure child and parents
• Short-acting β2-agonist via spacer (with face mask for those under 3), 2–4 puffs, increasing by 2 puffs every 2 min to 10 puffs if required
• Oral prednisolone 1–2 mg/kg, maximum 40 mg
• Monitor response for 15–30 min
• Short-acting β2-agonist via spacer, 10 puffs or nebulized (2.5 mg salbutamol in <8 years, 5 mg in >8 years), assess response and repeat as required
• Oral prednisolone or IV hydrocortisone
Consider:
• Inhaled ipratropium
• Nebulized magnesium
• IV β2-agonist (salbutamol) or aminophylline or magnesium
• Short-acting β2-agonist nebulized (2.5 mg salbutamol in <8 years, 5 mg in >8 years), assess response continuously and repeat as required (back to back)
• Oral prednisolone or IV hydrocortisone
• Nebulized ipratropium
Consider:
• Nebulized magnesium
• IV β2-agonist (salbutamol) or aminophylline or magnesium
Discuss with PICU
Assess response to treatment
Responding:
• Continue bronchodilators 1–4 h prn
• Discharge when stable on 4-h treatment
• Continue oral prednisolone for 3–7 days
At discharge:
• Review medication and inhaler technique
• Provide personalized asthma action plan
• Arrange appropriate follow-up
Not responding:
• Transfer to HDU/PICU
• Ensure senior medical review
• Consider IV therapies not already used (magnesium, aminophylline, β2-agonist)
• Consider CXR (to check for pneumothorax or infection) and blood gases
• Consider need for mechanical ventilation

Important notes

1–Magnesium sulphate is usually the first–line intravenous agent, but can cause profound hypotension.

2–both intravenous salbutamol and aminophylline, a loading dose is given over 20 minutes, followed by continuous infusion during which ECG(arrhythmia !!!)and blood electrolytes should be monitored.

3–Nausea and vomiting are common with aminophylline.

4–Intravenous salbutamol can cause tachycar–dia and lactic acidosis.

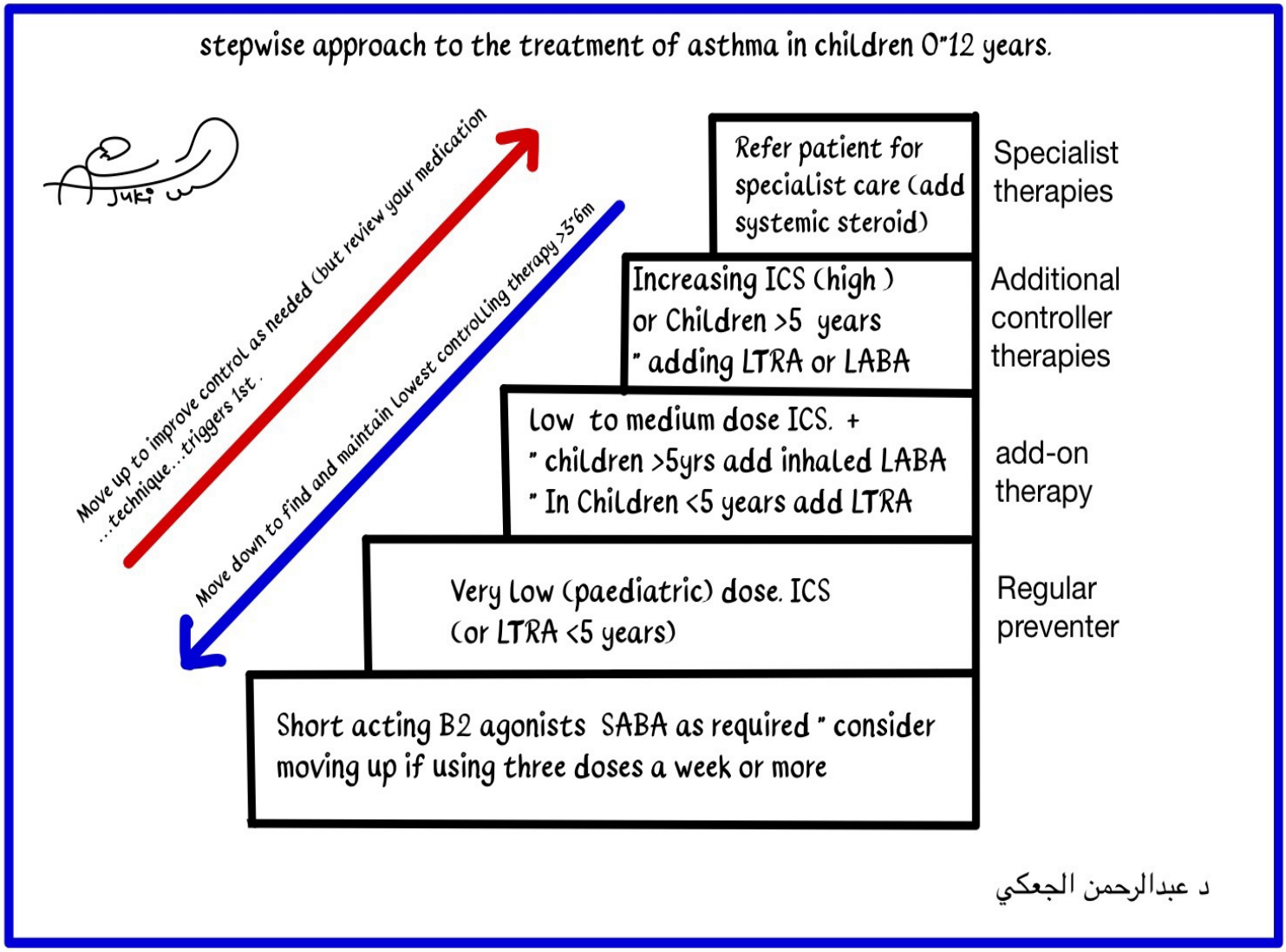
5–Antibiotics are only given if there are clinical features of bacterial infection.

Criteria for admission to hospital

1. have not responded adequately clinically, i.e. there is persisting breathlessness or tachypnoea after high–dose inhaled bronchodilator therapy
2. are becoming exhausted
3. still have a marked reduction in their predicted peak flow rate or FEV1 (<50%)
4. have a reduced oxygen saturation (<92% in air)

Classification of chronic asthma severity

	Mild intermittent	Mild persistent	Moderate persistent	Severe persistent
Days with symptoms	≤ 2 days/week	> 2 days /week	Daily	Throughout the day
Nighttime awakening	≤ 2 times/month	3–4 times /month	> 1time / week but not nightly	often 7times /week
PEFR & FEV_1	> 80%	> 80%	60–80%	< 60%
Interference with normal activity	None	some limitation	minor limitation	extremely limitation



Management of bronchial asthma according to assessment (Stepwise prophylaxis medication)

Drugs delivery devices in Br asthma (Choosing the correct inhaler)

1–Pressurized metered dose inhaler and spacer (pMDI +spacer):

Appropriate for all age groups (if age <2 years—spacer with face mask) (if age ≥3 years—spacer with mouthpiece)

Advantages:

1. increases drug deposition to the lungs (effective)
2. reduces oropharyngeal deposition (Less oral candidiasis)
3. Less systemic side–effects.
4. Useful for acute asthma attacks when poor inspiratory effort may impair the use of inhalers direct to the mouth.

2–• Breath–actuated metered dose inhalers [MDI]

(e.g. Autohaler): Appropriate for children >5 years. Less coordination needed than a pressurized metered dose inhaler without spacer. Useful for delivering β2–agonists.

3–• Dry powder inhaler –

from 4 years. Needs a good inspiratory flow,

therefore, less good severe asthma

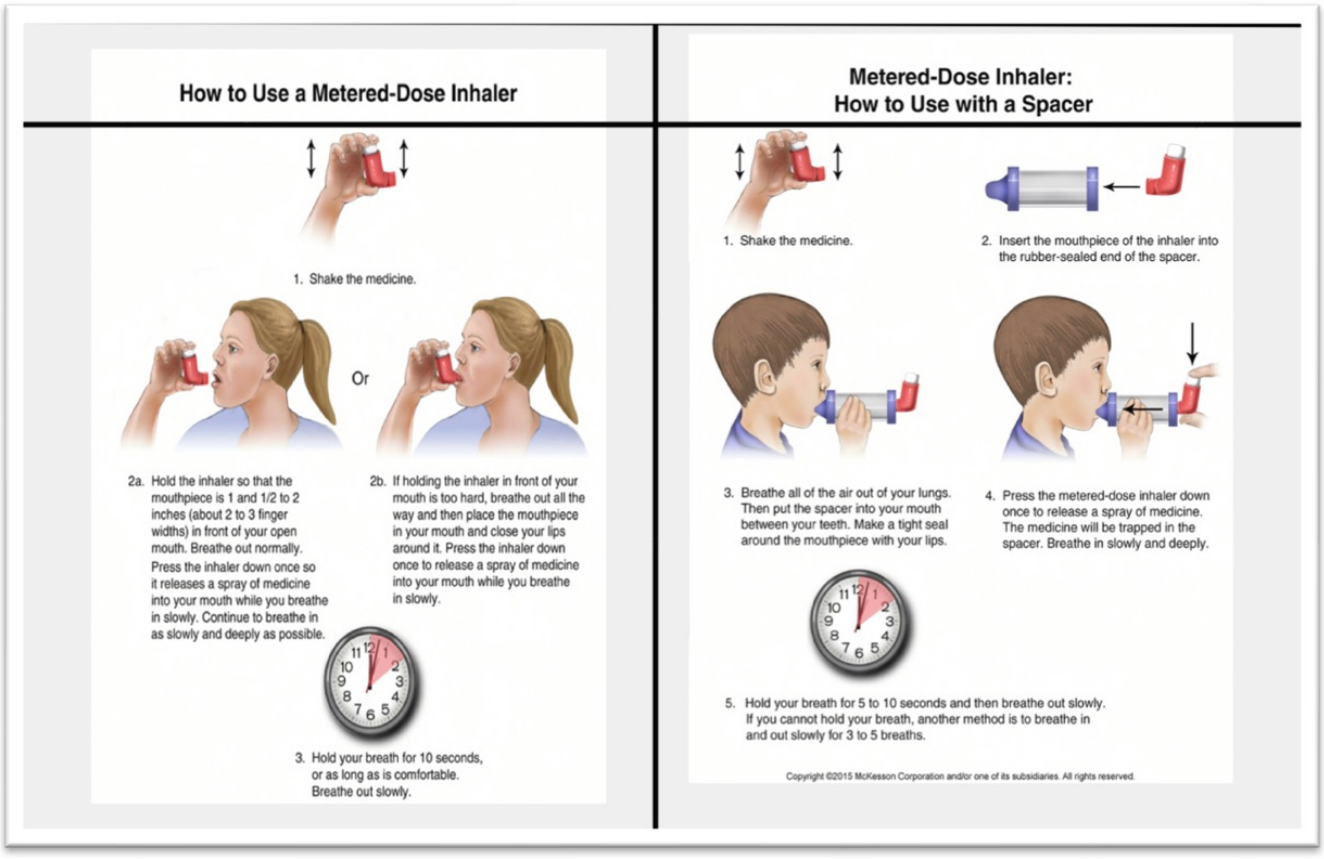
and during an asthma attack

4–• Nebulizer – any age.

A far less efficient method of drug delivery than metered–dose inhalers (MDI) plus spacer due to variation in droplet size.

Should always be administered using high–flow oxygen

not air. Only used in acute asthma where oxygen is needed.



Drugs used in management of bronchial asthma

B_2 agonist	Steroids	Leukotriene antagonist	Theophylline	Omalizumab
1-Short-acting β2-agonists SABA (rescue inhalers) A. Salbutamol B.Terbutaline *Rapid onset of action (10-15 min) *Effective for 2-4 hours *few side-effects. (Tachycardia-hypo K -Tremor)	**1-Systemic** 1. Hydrocortisone 2. Prednisone 3. Methyl prednisone 4.Dexamethasone *S/E Cushing syndrome	**Montelukast** *First choice of add-on therapy in children under 5 years. *Or addition of a LABA if not controlled	**Slow-release oral theophylline** (Methylxanthines) *Is an alternative add on therapy/ however, its use is limited in children by the high incidence of side-effects (vomiting, insomnia, headaches, poor concentration)	*Injectable monoclonal antibodies that act against IgE (**omalizumab**) and interleukin 5 (IL-5) (mepolizumab) are also available for selected children with severe atopic asthma.
2-long-acting β2-agonists LABA 1-Salmeterol 2-Formoterol *Effective for 12hours. *They are not used in acute asthma and should not be used without an inhaled corticosteroid * First choice of add-on therapy in a child over 5 years. * Are useful in exercise-induced asthma.	**2-Inhaler** 1-Budesonide 2-Beclometasone 3-Fluticasone 4-Mometasone *Called (preventers) are the most effective inhaled prophylactic drug therapy *They have no clinically significant side-effects when given in low dose.	**MART therapy** (combined ICS and LABA inhalers) as **m**aintenance **a**nd **r**eliever **t**herapy (MART) *Little evidence that this approach is effective in children and young people.		**Ipratropium bromide** *Anticholinergic broncho- dilator, is sometimes given to young infants when other broncho-dilators are found to be ineffective or in the treatment of severe acute asthma.

Drugs in asthma

Type of drug	Drug
Bronchodilators	
Short-acting β_2-agonists (rescue inhalers)	Salbutamol Terbutaline
Long-acting β_2-agonists (LABA)	Salmeterol Formoterol
Anticholinergic bronchodilator	Ipratropium bromide
Preventer therapy	
Inhaled steroids	Budesonide Beclometasone Fluticasone Mometasone
Methylxanthines	Theophylline
Leukotriene receptor antagonists (LTRA)	Montelukast
Oral steroids	Prednisolone
Anti-IgE monoclonal antibody	Omalizumab
Anti-IL5 monoclonal antibody	Mepolizumab

DD of wheezy chest in children

1– 1st attack wheeze	2. Recurrent attacks	3. Persistence wheeze
1. Acute bronchiolitis 2. FBA	1. Br. Asthma 2. GERD 3. Cystic fibrosis 4. Bronchiectasis 5. Viral episodic wheeze 6. Multiple trigger wheeze 7. Recurrent anaphylaxis (e.g. in food allergy , CMPI) Chronic aspiration 8. Bronchopulmonary dysplasia & Bronchiolitis obliterans 9. Tracheo–bronchomalacia 10. Immunodeficiency	1. Cystic fibrosis 2. Bronchiectasis 3. FBA 4. Vascular ring

The preschool child

Diagnosing asthma in preschool children is often difficult. Approximately half of all children wheeze at some time during the first 3 years of life. There are two main patterns of wheezing:

1–viral episodic wheezing 2– multiple trigger wheeze.

viral episodic wheezing	multiple trigger wheeze
• Triggers by viral URTIs	• Triggered by many stimuli which can include viral infections, cold air, dust, animal dander and exercise. • Associated with atopy, eczema, food allergy or family history of asthma.
• Due to abnormal immune response to viral infection causing inflammation and obstruction of the **small** airways. • Risk factors (smoking during and/or after pregnancy, prematurity and male sex) • It usually resolves by 5 years of age as increased airway size.	• positive family history of atopy • increases the likelihood the child has asthma and will benefit from asthma treatment.

CYSTIC FIBROSIS

Cystic fibrosis is Autosomal **recessive** genetic disorder

- CF is the **commonest life– limiting** inherited condition in Caucasians. It has an incidence of 1 in 2500 Live births and carrier rate of 1 in 25.
- CF is **multi systemic disease**
- CF caused by defect in gene called CF transmembrane conductance regulator (**CFTR**).Located in chromosome "7"
- Different gene mutation (≃ 2000)have been discovered – **ΔF 508** deletion is the most common 75%

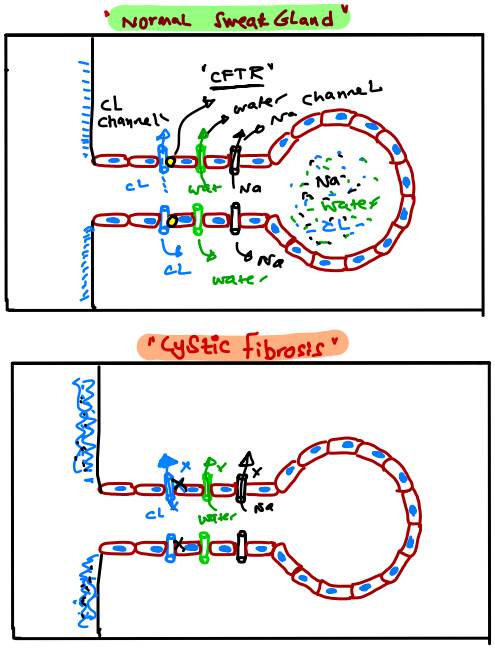
Pathophysiology:

CFTR gene responsible for production of CFTR protein that regulate the movement of chloride and sodium ions across epithelium membrane such as the alveolar epithelium of the lungs – leading to **thick and viscous secretion**

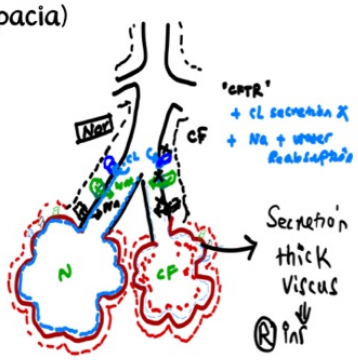
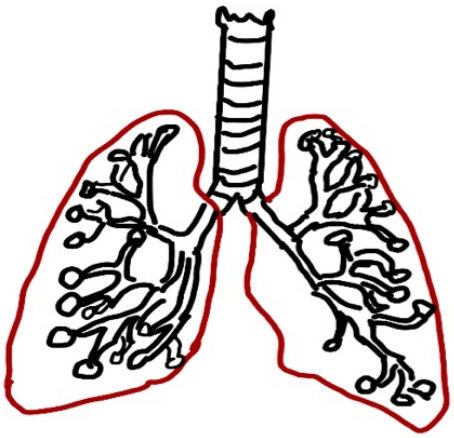
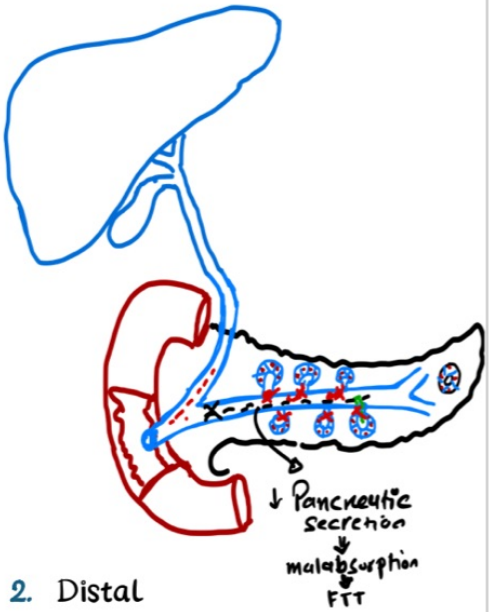
Clinical pictures: CF is **multi systemic disease** affect (Lung–pancreas–Liver–intestine)

In neonate:

- **Meconium ileus: 10–20%**
  Thick viscous Meconium – block the ilium – obstruction
  C/P: Failure to pass meconium in the 1st 48 hr of life
  Vomiting and abdominal distension
- Salty taste (the mother say that when she kiss the baby)
- Pseudo bartter syndrome – {hyop K, hyop cl, metabolic alkalosis}
- Prolonged Neonatal jaundice



In infancy and early childhood:

Respiratory ***	GIT**	others *
1. Recurrent chest infection: **early**: (Staph aureus,Hib) **late**: (Pseudomonas, Burkholderia cepacia)  2. Recurrent wheezy chest . 3. Nasal polyps 4. Sinusitis 5. Chronic infection → lung damage and bronchiectasis.  6. haemoptysis and pneumothorax may occur .	1. Pancreatic is insufficiency a. Exocrine function – malabsorption (Steatorrhea and FTT) b. Endocrine function – Diabetes mellitus (older child) called CF –related diabetes (CFRD)  2. Distal intestinal obstruction 3. Liver disease progress to Cirrhosis and portal hypertension (older child) 4. Rectal prolapse	1. GUS – infertility in males due to absence of the vas deferens. 2. Joints – Arthritis 3. Hands – Clubbing.

Cystic fibrosis should be considered in any child with recurrent infections, loose stools or faltering growth.

Diagnosis:

1. **Newborn screening**: Elevated immunoreactive trypsinogen (IRT).
2. **Sweat test:** gold standard diagnostic procedure .

Sweating is stimulated by applying a low-voltage current to pilocarpine applied to the skin. The sweat is collected into a special capillary tube or absorbed onto a weighed piece of filter paper. A raised sweat chloride concentra- tion (>60 mmol/L) confirms the diagnosis.

Chloride Concentration for Infants (older than 6 months) children and adults	Result
0 - 39 mmol/L	Cystic fibrosis is unlikely
40 - 59 mmol/L	Intermediate
≥ 60 mmol/L	Indicative of cystic fibrosis

- False positives:
  - Hypothyroidism
  - Addison disease
  - Ectodermal dysplasia
  - Glycogen storage disease
  - Edema
  - Malnutrition
  - Lab error (evaporation or contamination of sample)
- False negatives:
  - Edema
  - Malnutrition
  - Some CF mutations
  - Sample diluted

3. Chest X Ray: may indicate pulmonary involvement

4. Stool examination: evidence of malabsorption (low fecal elastase).

5. Liver function test (LFT)

6. Pulmonary function test: early obstructive airway disease.

7. Antenatal diagnosis available (if positive F/H)

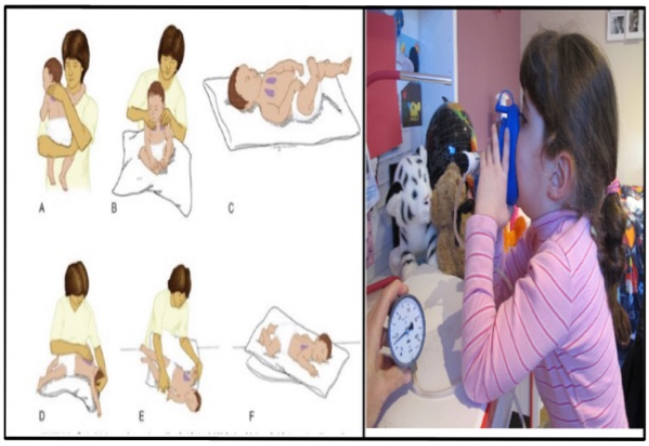
8. Genetic test (Confirmation of diagnosis can be made by testing for CFTR mutations).

Treatment: { CF is multisystemic disease – needs "multi disciplinary team" MDT}

1. Pulmonary care:

- Chest **physiotherapy** "at least twice a day"

(Chest percussion – postural drainage – deep breathing exercises)



Bronchodilators: Dilatation of airways easy movement of secretion

- **Antibiotics**:
- Oral during well periods "against staph aureus and pseudomonas
- IV for acute exacerbation (Flucloxacillin – Tobramycin or Ceftazidime)
- Nebulizer antipseudomonal antibiotic .
- **Mucolytics:** Nebulized mucolytics such as DNase and hypertonic saline or acetylcysteine 2 hr before physiotherapy it decrease the viscosity of secretions and the number of infective exacerbations

- Bilateral lung transplantation is the only therapeutic option for end-stage CF lung disease.

2. Nutrition

- High calorie diet
- Pancreatic enzyme replacement (**Creon**)
- Fat – soluble vitamin supplements

3. Rx of **complication** e.g[DM – Insulin]

4. Gene therapy – CFTR modulates (improve CF outcome)

Prognosis: The prognosis of CF improved (due to early diagnosis (neonatal screening) & new drugs , Expected survival age increased upto 35 years(now in some developed countries up to fifties).

(95% of patients with CF will die with **respiratory failure)**

Bronchiectasis

Definition: is abnormal irreversible dilatation of the bronchial tree.

Etiology:

1. Congenital	2. Acquired
• 1ry ciliary dyskinesia – AR as kartagener syndrome– Young syndrome • Cystic fibrosis "AR" • Hypogammaglobulinemia • Alpha antitrypsin "AAT" deficiency • Bronchopulmonary sequestration	• sever Measles • Whooping cough "pertussis" • Tuberculosis—HIV • Foreign body (missed) • Bronchiolitis obliterans (Adenovirus)

Clinical pictures:

Symptoms:

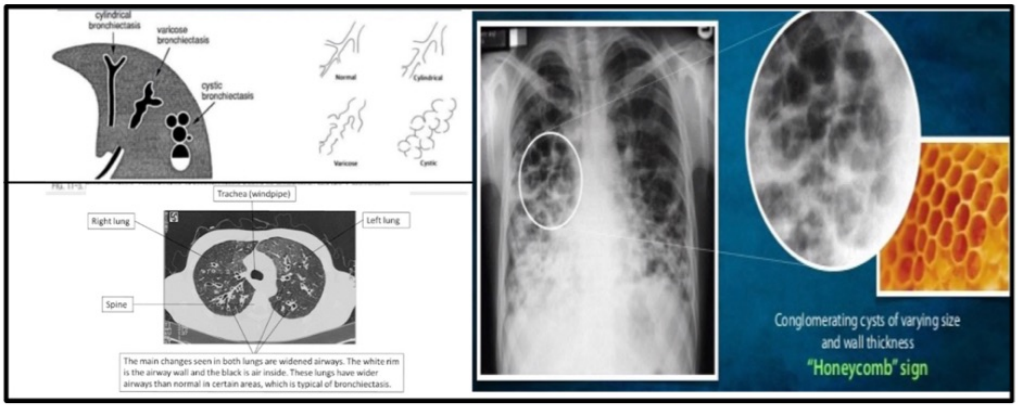
- **Chronic productive cough (large amount** of sputum)
- Hemoptysis (cough with blood)
- Recurrent pneumonia

Signs:

- Unwell child + FTT
- Clubbing
- Chest decreased air entry with crepitation Bil or unilateral

Diagnosis:

- Compatible history of chronic respiratory symptoms .
- Chest X Ray – early – normal and late – Honeycomb appearance.
- High resolution **CT– scan** – Diagnostic.
- Sputum analysis – for D_x infection.



Treatment:

1. Medical (main Rx) – Antibiotics and chest physiotherapy

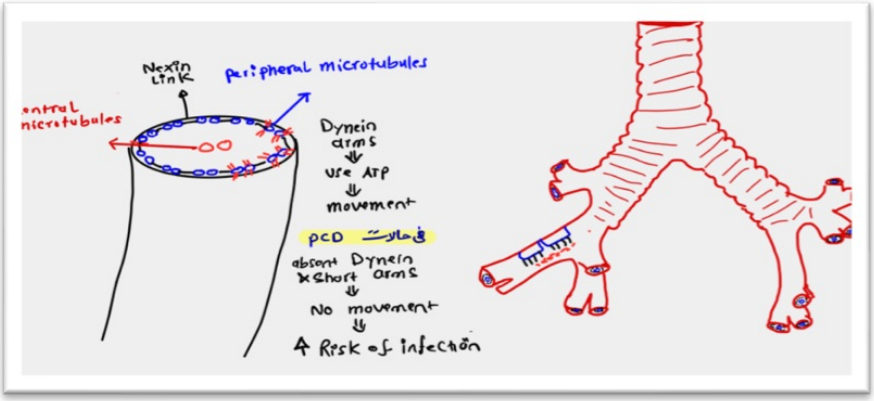
2. Surgical resection – In advanced complicated disease or local bronchiectasis or severe hemoptysis

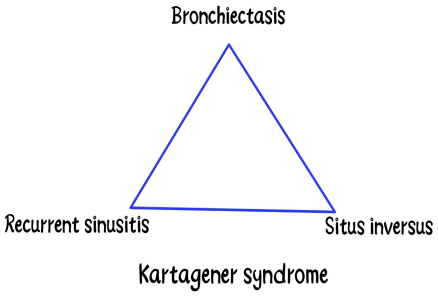
Complication:

1. Recurrent pneumonia, empyema, lung abscess
2. Respiratory failure
3. Severe hemoptysis
4. Amyloidosis

Primary ciliary dyskinesia

- Are Inherited disorders characterized by impaired ciliary function.
- Very rare –genetic disorders– 1 : 20000 (Autosomal recessive)





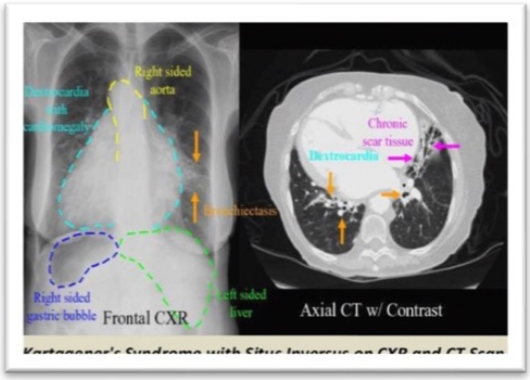
Kartagener syndrome

Clinical picture

Respiratory	Genitourinary	Orientation defects	CNS
• Neonate "RD" • Chronic cough • Recurrent pneumonia • Bronchiectasis • Chronic OM • Rhinitis, nasal polyp • Chronic sinusitis	• Immotile sperms (95% infertility in male) • Fallopian tubes defect (50% infertility in female)	• Situs inversus • Asplenia • CHD	• Hydrocephalus • Retinitis Pigmentosa

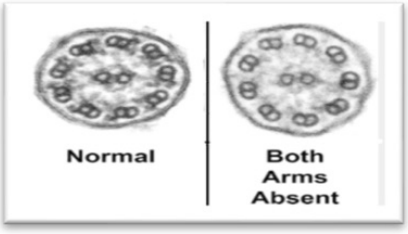
Diagnosis:

- CXR and CT chest
- CT sinus
- Electron microscopy and microscopy analysis of nasal sample
- Genetic testing
- Nasal nitric oxide as screening test



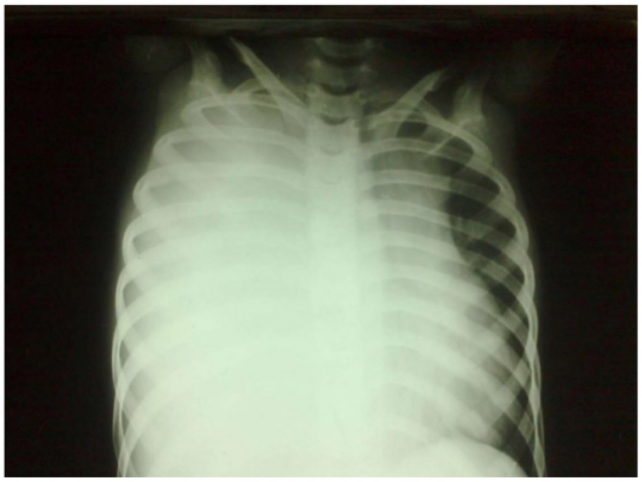
Treatment:

- Airway clearance therapy
- Bronchodilator – antibiotics
- ENT consultations



Pleural effusion

Definition: Means fluid in pleural cavity



Right massive pleural effusion

A. Transudative:

1. Nephrotic syndrome
2. Liver cirrhosis
3. Hypoproteinemia
4. Congestive heart failure

B. Exudative:

1. Pneumonia
2. TB
3. Ruptured lung abscess.tumors .
4. CTD as SLE

	Transudative PE	Exudative PE
Color	Clear	Turbid
Proteins	Low (<3 gm/dL)	High (> 3gm/dL)
Cells	Low	High (pmn)
Specific Gravity	Low	High
LDH	< 200 IU/L	> 200 IU/L
Organisms	No	±

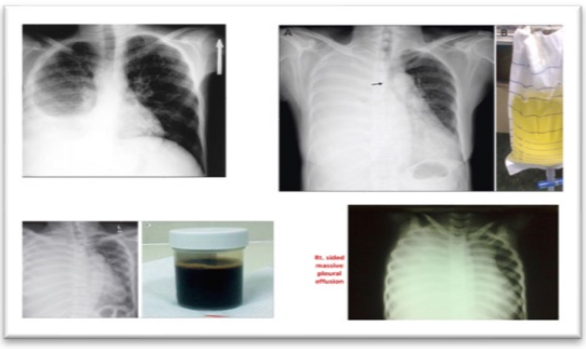
Clinical pictures:

- Breathlessness
- Chest pain, RD signs –cyanosis .
- Manifestation of the cause

On examination:

- Inspection – Decreased movement on affected side
- Palpation – Trachea shifted to opposite side, decrease chest expansion and TVF
- Percussion – Stony dullness
- Auscultation – Decreased air entry or absent on affected side

Ix

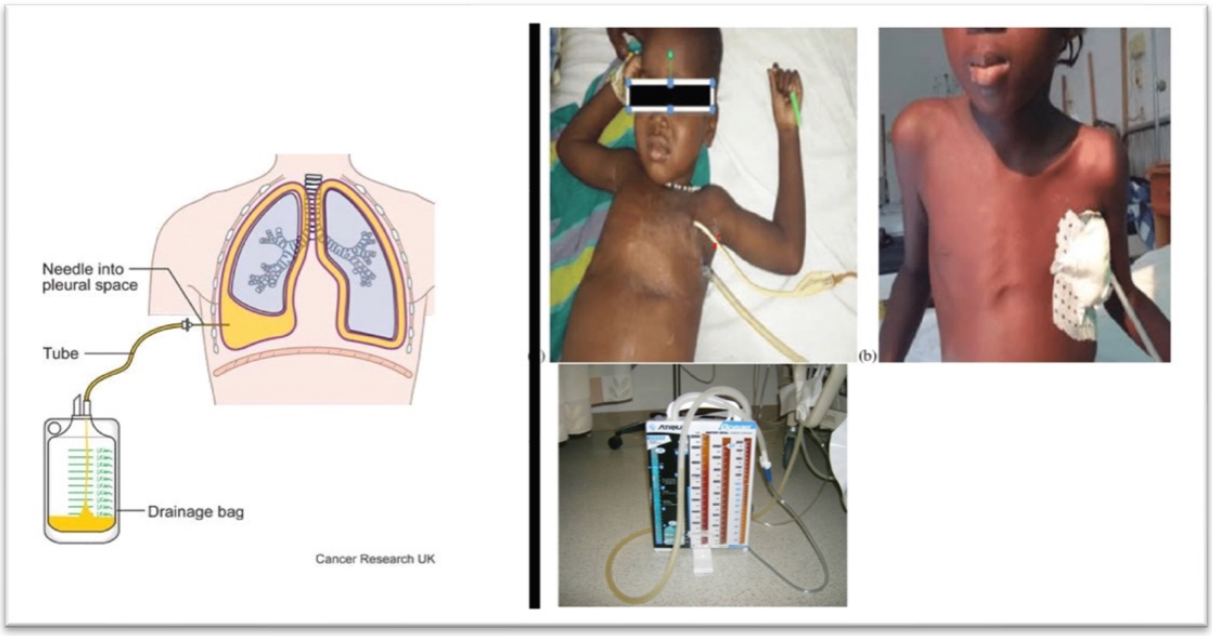


1. Chest X Ray
2. CT scan "chest"

Rx:

(According to severity and underlying cause)

1. Rx the underlying cause,
2. Thoracocentesis with or without chest tube underwater seal .



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تنبيه:

1. هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال ، كل ذلك كا ن ممزوج بحنكة العطاء الاكاديمي للمادة.
2. لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الآ بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي.